

## MATHEMATICS GRADE 10: SURFACE AREA AND VOLUME OF SOLIDS

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.4 (10 Lessons)*

| SUB-STRAND OVERVIEW |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| Grade Level         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Subject             | Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Strand              | Strand 2.0: Geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Sub-Strand          | Sub-Strand 2.7: Surface Area and Volume of Solids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Total Duration      | 10 lessons × 40 minutes = 400 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sub-Strand Content  | <ul style="list-style-type: none"> <li>– Surface area of prisms (triangular, rectangular, and other prisms)</li> <li>– Surface area of pyramids (triangular and rectangular-based pyramids)</li> <li>– Surface area of cones</li> <li>– Surface area of spheres</li> <li>– Surface area of frustums of a cone</li> <li>– Surface area of frustums of a pyramid</li> <li>– Surface area of composite solids</li> <li>– Volume of prisms (triangular, rectangular, and other prisms)</li> <li>– Volume of pyramids (triangular and rectangular-based pyramids)</li> <li>– Volume of cones</li> <li>– Volume of spheres</li> <li>– Volume of frustums of a cone</li> <li>– Volume of frustums of a pyramid</li> <li>– Volume of composite solids</li> <li>– Real-life applications of surface area and volume of solids</li> </ul>               |
| Learning Outcomes   | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>determine the surface area of prisms, pyramids, cones, and spheres in different situations,</li> <li>determine the surface area of frustums of a cone and a pyramid in different situations,</li> <li>calculate the surface area of composite solids in real-life contexts,</li> <li>calculate the volume of prisms, pyramids, cones, and spheres in different situations,</li> <li>determine the volume of frustums of a cone and a pyramid in different situations,</li> <li>determine the volume of composite solids in real-life contexts,</li> <li>apply surface area and volume of solids to solve real-life problems,</li> <li>appreciate the use of surface area and volume of solids in day-to-day activities.</li> </ol> |
| Core Competencies   | <ul style="list-style-type: none"> <li>– Communication and Collaboration: the learner discusses with peers and works in groups to derive and verify formulae for surface area and volume of various solids, sharing findings and explaining reasoning to classmates.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                                                   | <ul style="list-style-type: none"> <li>– Critical Thinking and Problem Solving: the learner applies knowledge of surface area and volume to solve real-life problems such as estimating the amount of material needed to construct a grain storage silo or a water tank in the Rift Valley.</li> <li>– Digital Literacy: the learner uses digital devices to access YouTube videos on surface area and volume of frustums and composite solids, and uses calculators to verify computed values.</li> <li>– Learning to Learn: the learner builds on prior knowledge of area and volume of basic shapes to extend understanding progressively to frustums and composite solids.</li> <li>– Self-efficacy: the learner develops confidence in applying mathematical formulae to multi-step problems involving composite solids encountered in everyday Kenyan contexts.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Core Values</b>                                | <ul style="list-style-type: none"> <li>– Responsibility: the learner takes care of mathematical instruments, calculators, and digital devices used during investigations of surface area and volume.</li> <li>– Respect: the learner listens to and values the contributions of peers during group discussions on the properties and measurements of solids.</li> <li>– Unity: the learner works collaboratively in groups to measure, compute, and verify surface areas and volumes of physical models of solids.</li> <li>– Social Justice: the learner recognises how accurate computation of surface area and volume contributes to fair and efficient use of resources, such as equitable distribution of grain storage facilities across counties in Kenya.</li> <li>– Integrity: the learner records and reports measurements and calculations honestly, without falsifying results during practical investigations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                      |
| <b>Science &amp; Engineering Practices</b>        | <ul style="list-style-type: none"> <li>– Developing and Using Models: learners construct and manipulate physical nets and three-dimensional models of prisms, pyramids, cones, and frustums to derive surface area formulae, then revise their models as understanding deepens.</li> <li>– Using Mathematics and Computational Thinking: learners apply geometric formulae, substitute values, and use calculators to compute surface areas and volumes accurately, comparing results across different solid types.</li> <li>– Analysing and Interpreting Data: learners measure physical objects (e.g., a cylindrical uji pot, a conical grain store, a rectangular water tank) and analyse whether computed values match real-world capacity labels.</li> <li>– Constructing Explanations: learners explain, using derived formulae and worked examples, why a frustum has a larger surface area than a cone of the same base and slant height, and how composite solids combine individual formulae.</li> <li>– Obtaining, Evaluating, and Communicating Information: learners watch curated YouTube videos on frustum surface area and volume, evaluate the methods demonstrated, and communicate their understanding through poster presentations and model demonstrations.</li> </ul> |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Life Skills (Creative Thinking): the learner designs a composite solid container (e.g., a water storage tank combining a cylinder and hemisphere) and computes its surface area and volume, applying creative thinking to a real Kenyan engineering context.</li> <li>– Life Skills (Problem Solving): the learner solves multi-step problems involving the surface area and volume of solids encountered in everyday situations, such as estimating the cost of painting a conical roof or filling a frustum-shaped grain silo in Kitale.</li> <li>– Environmental Education: the learner explores how efficient design of solids (minimising surface area for a given volume) contributes to sustainable use of materials, such as reducing the amount of metal sheet used to manufacture water tanks for rural communities in Turkana County.</li> <li>– Financial Literacy: the learner calculates the cost of materials needed to construct or cover solids, applying unit cost to computed surface areas to estimate project budgets relevant to small businesses and farming enterprises in Kenya.</li> </ul>                                                                                                                               |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Architecture and Construction: architects and civil engineers in Kenya use surface area and volume calculations to design buildings, estimate materials, and price construction projects such as grain silos, water tanks, and domed rooftops.</li> <li>– Manufacturing and Industrial Design: manufacturers of containers, packaging, and storage facilities (e.g., metal fabricators in</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                                                 | <p>Industrial Area, Nairobi) rely on surface area formulae to minimise material waste and optimise production costs.</p> <ul style="list-style-type: none"> <li>– Agriculture and Agribusiness: agronomists and farmers in the Rift Valley calculate the volume of conical or cylindrical grain stores and water reservoirs to manage harvests and irrigation efficiently.</li> <li>– Medicine and Health Sciences: medical professionals and biomedical engineers use volume calculations of spherical and cylindrical vessels when designing pharmaceutical containers, IV drip bags, and medical implants.</li> <li>– Water Engineering and Environmental Science: water engineers designing dams, tanks, and pipelines for counties such as Makueni and Baringo apply volume and surface area knowledge to ensure adequate water storage and distribution.</li> <li>– Art and Graphic Design: Kenyan artists and product designers use geometric solid properties when creating sculptures, jewellery moulds, and product packaging.</li> </ul>                                                                                                                                                                                       |
| <b>Focus for Lessons</b>                        | <p>This unit focuses on equipping Grade 10 learners with the knowledge and skills to compute the surface area and volume of a wide range of three-dimensional solids — including prisms, pyramids, cones, spheres, frustums, and composite solids — and to apply these computations in authentic Kenyan real-life contexts such as designing grain stores, water tanks, and roofing structures. The unit progresses systematically from familiar solids (prisms and pyramids) through curved surfaces (cones and spheres) to more complex shapes (frustums and composite solids), building conceptual understanding at each stage. A phenomenon-driven, collaborative approach is used throughout, with learners constructing physical nets, watching curated video demonstrations, and solving contextualised problems to develop both procedural fluency and deep conceptual understanding.</p>                                                                                                                                                                                                                                                                                                                                         |
| <b>Driving Question / Key Inquiry Questions</b> | <p>DRIVING QUESTION: How does the shape of a solid determine how much material is needed to build it and how much it can hold — and how do engineers and manufacturers in Kenya use this knowledge to design efficient, cost-effective structures?</p> <p>KICD KEY INQUIRY QUESTIONS:</p> <ol style="list-style-type: none"> <li>1. How do we apply surface area and volume of solids in day-to-day activities?</li> <li>2. How does the shape of a solid affect its surface area and volume?</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Anchoring Phenomenon</b>                     | <p>ANCHORING PHENOMENON: A water harvesting tank manufacturer in Kisumu produces two types of storage tanks for rural households in Siaya County: a cylindrical tank topped with a hemispherical lid, and a frustum-shaped tank (wider at the bottom, narrower at the top). Both tanks are advertised as holding exactly 5,000 litres of water. When a group of Grade 10 students visits the factory, they notice that the frustum tank uses noticeably less metal sheet than the composite cylinder-hemisphere tank, yet holds the same volume of water. The students are puzzled: if both tanks hold the same amount of water (same volume), why does one require significantly less material to build (smaller surface area)? This observation drives the learners to investigate how the shape of a solid determines its surface area independently of its volume, and how manufacturers and engineers in Kenya strategically choose solid shapes to minimise cost while maximising storage capacity.</p>                                                                                                                                                                                                                             |
| <b>Storyline Thread</b>                         | <p>Lesson 1 — Surface Area of Prisms and Pyramids: Learners are introduced to the anchoring phenomenon (the Kisumu water tank factory) and make initial predictions about why the two tanks use different amounts of metal. They then derive and apply formulae for the surface area of triangular and rectangular prisms and pyramids using physical nets, grounding all calculations in the context of flat metal sheets being cut and folded at the factory.</p> <p>Lesson 2 — Surface Area of Cones and Spheres: Learners extend their investigation to curved solids, deriving the lateral and total surface area formulae for cones and spheres. They connect this to the supporting phenomenon of spherical mandazi and conical roof turrets in Nairobi, reinforcing why curved surfaces have distinct formulae from flat-faced solids.</p> <p>Lesson 3 — Surface Area of Frustums of a Cone: Using the YouTube video on the surface area of a frustum of a cone, learners investigate how cutting a cone parallel to its base produces a frustum. They derive the surface area formula for a cone frustum and apply it to the anchoring phenomenon — calculating the surface area of the frustum-shaped water tank in Kisumu.</p> |

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|                             | <p>Lesson 4 — Surface Area of Frustums of a Pyramid: Building on Lesson 3, learners watch the YouTube video on the surface area of a frustum of a pyramid and derive its formula. They apply this to contextualised problems such as calculating the metal sheet needed for a frustum-shaped grain store in Kitale, updating the Driving Question Board with new understanding.</p> <p>Lesson 5 — Surface Area of Composite Solids: Learners watch the YouTube video on calculating the surface area of composite solids and apply a systematic strategy (identify components, subtract hidden faces, sum exposed surfaces) to compute the surface area of composite solids. They apply this to the Kericho tea canister and the cylinder-hemisphere water tank from the anchoring phenomenon, directly comparing it to the frustum tank from Lessons 3–4.</p> <p>Lesson 6 — Volume of Prisms, Pyramids, Cones, and Spheres: Learners shift focus from surface area to volume, deriving and applying formulae for the volume of prisms, pyramids, cones, and spheres. Using the Kitale maize storage supporting phenomenon, they compute and compare volumes of rectangular and conical grain stores, explaining the farmer's observation about differing storage capacities.</p> <p>Lesson 7 — Volume of Frustums of a Cone: Using the YouTube video on the volume of a frustum of a cone, learners derive the frustum volume formula from the difference of two cones. They calculate the volume of the frustum-shaped water tank from the anchoring phenomenon, confirming whether it truly holds 5,000 litres, and update the DQB to track the growing resolution of the driving question.</p> <p>Lesson 8 — Volume of Frustums of a Pyramid: Building on Lesson 7, learners watch the YouTube video on the volume of a frustum of a pyramid and derive its formula. They apply it to problems involving pyramid-frustum grain stores and architectural structures, connecting to careers in construction and agribusiness in Kenya.</p> <p>Lesson 9 — Volume of Composite Solids: Using the YouTube video on the volume of composite solids, learners apply a decomposition strategy to compute volumes of composite solids (e.g., cylinder + hemisphere water tank, cone + cylinder silo). They revisit the anchoring phenomenon in full — computing both the surface area and volume of both tank designs side by side — to finally explain why the frustum tank uses less material while holding the same volume.</p> <p>Lesson 10 — Real-Life Applications and Model Building: Learners consolidate their understanding by solving a rich real-life task set in a Kenyan context (e.g., designing the most cost-efficient water storage tank for a school in Siaya County). They present their final annotated cumulative models, update the Driving Question Board with complete explanations, and reflect on how surface area and volume of solids are used across careers in architecture, agriculture, manufacturing, and water engineering in Kenya.</p> |
| <b>Supporting Phenomena</b> | <ul style="list-style-type: none"> <li>– A mandazi vendor in Mombasa's Old Town deep-fries mandazi dough cut into different shapes — spherical balls, conical rolls, and flat triangular prisms — and notices that the spherical mandazi browns more evenly and absorbs less oil. Learners investigate how the surface-area-to-volume ratio of each shape explains the frying behaviour and oil absorption.</li> <li>– A farmer in Kitale stores harvested maize in two grain stores: one is a rectangular block structure and the other is a conical thatched silo of the same height. The conical silo holds less grain than expected. Learners compute and compare the volumes of both structures to explain the difference in storage capacity.</li> <li>– A Kenyan artisan in Kericho crafts tea canister gift sets by soldering flat metal sheets into composite solid shapes (cylinder + cone lid). Learners are given a flat net of the canister and must calculate the exact area of metal sheet required — connecting the net (2D surface) to total surface area (3D solid).</li> <li>– Workers at a construction site in Nairobi's Westlands are painting a roof consisting of four conical turrets and a central</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|  | hemispherical dome. The foreman needs to purchase the correct amount of paint. Learners calculate the lateral surface area of each conical turret and the curved surface area of the dome to determine the total area to be painted and the litres of paint required. |
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## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (80 minutes): Surface Area of Prisms and Pyramids — Launching the Phenomenon

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>Purpose</b>              | To introduce learners to the anchoring phenomenon of the Kisumu water tank factory and establish foundational understanding of surface area by deriving and applying formulae for prisms and pyramids through physical nets.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Knowledge</b>            | By the end of this lesson, the learner should be able to: (a) define surface area as the total area of all faces of a 3D solid; (b) identify and sketch nets of triangular prisms, rectangular prisms, and pyramids; (c) derive surface area formulae for triangular prisms, rectangular prisms, and square-based pyramids; (d) calculate the surface area of prisms and pyramids in real-world contexts.                                                                                                                                                                                                                               |
| <b>Skills</b>               | The learner should be able to: (a) construct and interpret nets of 3D solids to determine total surface area; (b) apply the correct surface area formula to solve problems related to material estimation; (c) communicate mathematical reasoning clearly using correct terminology.                                                                                                                                                                                                                                                                                                                                                    |
| <b>Attitudes</b>            | The learner should: (a) appreciate how geometric reasoning supports efficient, cost-effective engineering and manufacturing decisions in Kenya; (b) show curiosity and persistence when investigating why different shapes use different amounts of material for the same volume.                                                                                                                                                                                                                                                                                                                                                       |
| <b>Key Inquiry Question</b> | How do we apply surface area and volume of solids in day-to-day activities?   How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Purpose in Storyline</b> | This is the opening lesson of the sub-strand. Learners are introduced to the anchoring phenomenon — a water tank factory in Kisumu producing a cylindrical-hemispherical tank and a frustum tank, both holding 5,000 litres, but using different amounts of metal sheet. Learners make initial predictions about the phenomenon, then build foundational surface area skills using prisms and pyramids modelled as flat metal sheets being cut and folded at the factory. This lesson opens the Driving Question Board and positions learners to investigate how shape determines surface area independently of volume across the unit. |
| <b>Safety Notes</b>         | Learners using scissors to cut nets should handle them with care and pass them blade-down. No other significant hazards anticipated. Ensure adequate desk space for net construction activities.                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### B. LESSON OVERVIEW

Learners open the sub-strand by encountering the anchoring phenomenon: a water tank factory in Kisumu manufactures two tanks — a composite cylinder-hemisphere and a frustum — that both hold exactly 5,000 litres, but the frustum uses noticeably less metal sheet. After recording initial predictions, learners connect this puzzle to surface area. They explore physical nets of rectangular prisms, triangular prisms, and square-based pyramids — modelled as flat metal sheets being cut and folded — to derive surface area formulae. Learners apply these formulae to factory-relevant problems, update their initial predictions, and co-create the opening Driving Question Board. The lesson closes with a cumulative class model that maps 'shape → net → surface area formula' as the first layer of the unit's conceptual model.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                     | Learner Experience                                                                         | Teacher Moves                                                               | Sensemaking Strategy                                                                                  | Formative Assessment Strategy                                                              |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Surface area</a> | Learners are shown a large photograph (or teacher-drawn diagram on the board) of two tanks | Display the tank diagram/photograph prominently.<br>Read the scenario aloud | Predict-before-instruction (Eliciting Prior Knowledge): By committing predictions to paper before any | Observable indicator: Every learner has written at least 3 sentences and a sketch in their |

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| <p><a href="#">using a net: triangular prism</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Apply volume of solids</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar readings</p>                                            | <p>side by side at a factory in Kisumu: Tank A is a tall cylinder with a hemispherical dome on top; Tank B is a frustum (wide base, narrower top). A label on each reads: 'Capacity: 5,000 litres.' A factory worker speech bubble says: 'Tank B always uses less metal sheet than Tank A — but both hold the same water!' Learners individually write a 3-sentence prediction in their exercise books answering: 'Why do you think Tank B uses less metal sheet than Tank A even though both hold 5,000 litres?' They annotate their prediction with a rough sketch. After 4 minutes of silent individual writing, learners share predictions with a shoulder partner for 2 minutes. Four learners are cold-called to share with the class.</p>            | <p>dramatically: 'Imagine you work at this factory in Kisumu. Your manager tells you — Tank B always needs less metal. But they hold exactly the same water. Why?' Allow 10–15 seconds of WAIT TIME after posing the question before learners begin writing. Circulate and read 3–4 predictions silently to identify a range of ideas (surface-area reasoning, shape reasoning, misconceptions about volume). Cold-call 4 learners using name cards: choose one who mentions 'shape', one who mentions 'size', one who mentions 'volume', and one with a misconception. Record each prediction verbatim on the board under the heading 'OUR PREDICTIONS — Lesson 1.' Say: 'We are NOT confirming or correcting these yet. We will return to them at the end of today and at the end of the unit.'</p> | <p>instruction, learners activate prior knowledge about shape and material, create a cognitive anchor for new content, and generate personal investment in resolving the puzzle. Recording predictions publicly on the board creates accountability and a reference point for revision throughout the unit.</p>                                                                                                             | <p>book within 4 minutes. Teacher scans books during circulation — look for whether learners distinguish between 'how much water a tank holds' (volume) and 'how much metal is used' (surface area). Flag learners who conflate the two concepts — these learners need targeted scaffolding in the Explain phase. Cold-call responses reveal the range of prior conceptions to inform pacing.</p>                                                                                                                                                                        |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Surface area using a net: rectangular prism</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/><a href="#">Surface Area of Triangular Prisms (Flexbook)</a><br/>Source: CK-12<br/><a href="#">Search ARES</a> for similar readings</p> | <p>Learners work in groups of 4. Each group receives a pre-printed net cut-out kit containing: (a) a net of a rectangular prism (cuboid) labelled with dimensions representing a simple rectangular metal water trough — length 80 cm, width 40 cm, height 40 cm; (b) a net of a triangular prism representing a triangular roof support beam used in tank assembly — triangular face base 30 cm, height 20 cm, length 60 cm; (c) a net of a square-based pyramid representing a decorative roof cap — base 40 cm × 40 cm, slant height 35 cm. Each group also has a plain A3 sheet, ruler, and scissors. Task 1 (10 min): Learners fold each net into its 3D solid and observe: 'What faces does each solid have? How many? What shape are they?' They</p> | <p>Distribute net kits to groups before the phase begins to save time. Give clear verbal instructions: 'Step 1 — fold each net. Step 2 — observe the faces. Step 3 — unfold and calculate total area face by face. Do NOT use a formula yet — use only what you know about rectangles and triangles.' Set a visible timer on the board for each task. Circulate actively every 90 seconds. Ask probing questions to groups: 'How many rectangular faces does your prism have?', 'Are any faces identical? How does that help you?', 'If this net were a real sheet of metal at the Kisumu factory, what would cutting along these lines mean?' After Task 2, ask one group to present their rectangular prism working on the board. Allow 10–15 seconds of</p>                                        | <p>Net Construction and Face Inventory (Concrete-to-Abstract Bridge): Physically folding and unfolding nets connects the abstract idea of 'total surface area' to the tangible factory action of cutting and folding metal sheets. Summing face areas before seeing a formula builds procedural understanding from first principles, ensuring the formula is later recognised as a shortcut rather than a mystery rule.</p> | <p>Observable indicators: (1) Each group's observation table is complete with correct face counts (cuboid = 6, triangular prism = 5, square pyramid = 5). (2) Group surface area totals for the rectangular prism are within 5% of the correct value: <math>2(80 \times 40 + 80 \times 40 + 40 \times 40) = 2(3200 + 3200 + 1600) = 16,000 \text{ cm}^2</math>. (3) Teacher checks that learners are summing ALL faces (a common error is omitting the base). Prompt any group missing the base: 'In the factory, does the bottom of the tank need metal sheet too?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | complete an observation table in their books: Solid   Number of faces   Shapes of faces   Sketch of net. Task 2 (8 min): Learners unfold each solid back to its net and, WITHOUT any formula, physically count and measure the area of each face by subdividing into known shapes (rectangles, triangles), then sum all face areas to estimate total surface area. They record their totals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | WAIT TIME before cold-calling if no group volunteers. Cold-call 2 groups to compare totals — prompt: 'Did both groups get the same answer? If not, what might have caused the difference?'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Explain Phase</b><br> <b>VIDEO:</b><br><a href="#">Recognizing common 3D shapes</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Surface Area of Triangular Prisms Practice</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | The class reconvenes. The teacher guides learners to generalise their face-counting into formal surface area formulae through structured questioning, using the factory context throughout. Part A — Rectangular Prism (Cuboid): Learners are guided to express: $SA = 2(lw + lh + wh)$ . They apply this to find the metal needed to make an open-topped rectangular water storage trough at the Kisumu factory (open top means subtract one $lw$ face). Part B — Triangular Prism: Learners derive $SA = 2(\frac{1}{2} \times b \times h) + (\text{perimeter of triangle} \times \text{length}) = bh + (a + b + c)l$ , connecting each term to a named face of the net. Part C — Square-Based Pyramid: Learners derive $SA = l^2 + 4 \times (\frac{1}{2} \times l \times s) = l^2 + 2ls$ , where $l$ = base side, $s$ = slant height. They apply this to find the metal needed for a decorative pyramid roof cap (base side 40 cm, slant height 35 cm). Learners then revisit their Predict Phase predictions and individually annotate: 'I now know that surface area means... My original prediction was... I would now change/keep it because...' | Use the board to build each formula incrementally — write each face's area expression, then combine. Use colour-coding: rectangular faces in blue, triangular faces in red, square base in green. Ask at each step: 'Which face of the net does this term represent? Point to it on your unfolded net.' Allow 10–15 seconds WAIT TIME. Cold-call 3 learners per formula derivation. Explicitly connect each formula to the factory context: 'If you are a metal sheet cutter at the Kisumu factory, this term — $2lw$ — tells you what?' After the pyramid formula, pose: 'The factory makes 200 pyramid roof caps per week. Each one needs how many square centimetres of metal sheet? How many square metres is that?' (Answer: $200 \times (40^2 + 2 \times 40 \times 35) = 200 \times (1600 + 2800) = 200 \times 4400 = 880,000 \text{ cm}^2 = 88 \text{ m}^2$ .) Give 3 minutes for individual working, then cold-call 2 learners to present. For the prediction-annotation task, give 3 minutes silent individual writing. Say: 'Write in your own words — no group discussion yet.' | Incremental Formula Derivation with Contextual Anchoring: Each algebraic term in the formula is traced back to a physical face of the factory metal sheet. This prevents rote memorisation and builds transferable understanding. The prediction-annotation task (I used to think... now I think...) makes conceptual change visible and metacognitive, reinforcing that science and mathematics learning is a process of refining ideas with evidence. | Observable indicators: (1) Learners can correctly identify which term in each formula corresponds to which face — check by asking 3 cold-called learners to point to the net face as they name the term. (2) The pyramid roof cap problem is solved correctly by at least 80% of learners working individually (teacher circulates and checks books). (3) Prediction annotations show movement from vague 'shape-based' reasoning toward precise 'surface area formula' reasoning. Flag learners whose annotations still conflate surface area and volume — provide a direct prompt: 'Surface area is about the outside skin. Volume is about what fits inside. Are they the same?' |
| <b>Driving Question</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Each learner receives two sticky notes (or writes in a designated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Distribute sticky notes or direct learners to a dedicated page in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Driving Question Board — Know/Wonder Protocol: The DQB                                                                                                                                                                                                                                                                                                                                                                                                  | Observable indicators: (1) Every learner has posted or recorded                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| <p><b>Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area of a box (cuboid)</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface area word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>DQB section of their book if physical sticky notes are unavailable). On sticky note 1 (WHAT WE KNOW — green): learners write one thing they are now confident about from today's lesson connected to the main driving question. On sticky note 2 (WHAT WE STILL WONDER — yellow): learners write one question that today's lesson raised or left unanswered about the Kisumu tank phenomenon. Learners post their notes on the class Driving Question Board (a section of the classroom wall or a large sheet of manila paper headed: 'HOW DOES SHAPE DETERMINE HOW MUCH MATERIAL IS NEEDED AND HOW MUCH A SOLID CAN HOLD?'). The teacher reads 5–6 posted questions aloud and clusters them into emerging themes: (a) questions about cylinders/hemispheres, (b) questions about frustums, (c) questions about volume vs. surface area relationships, (d) questions about cost. The teacher says: 'These are our investigation questions for the unit. Each lesson will help us answer more of them.'</p> | <p>their books. Give exactly 3 minutes for silent writing. Circulate to ensure all learners write a genuine question on their yellow note — prompt any learner who is stuck: 'Think about Tank A and Tank B. What do you still not understand about why they use different amounts of metal?' Read selected questions aloud with enthusiasm: 'Ooh — someone is asking whether a sphere would use even LESS metal for 5,000 litres. That is a fantastic question — we will get to that!' Physically cluster sticky notes into themes on the board using a marker to draw circles around clusters and label them. Photograph the DQB for the class record. Say: 'This board will grow every lesson. By the end of this sub-strand, we will have answered every single cluster.'</p> | <p>externalises and organises the class's collective growing understanding. Separating 'know' from 'wonder' makes epistemic progress visible over time and gives learners agency over the direction of investigation. Clustering questions teaches learners to see conceptual relationships between sub-questions, mirroring how professional engineers and scientists organise complex problems.</p>                                                  | <p>both a 'know' and a 'wonder' entry. (2) 'Know' entries reference surface area vocabulary (net, face, formula, prism, pyramid) rather than vague statements. (3) 'Wonder' entries show genuine engagement with the phenomenon — questions about cylinders, frustums, volume, cost, or material efficiency indicate readiness for upcoming lessons. (4) Teacher notes which concept clusters attract the most 'wonder' questions — this informs differentiation and pacing in Lessons 2–5.</p> |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area using a net: triangular prism</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Surface Area of</a></p>                                                                                                                                                                                               | <p>Learners individually draw the first version of their Cumulative Unit Model in their books. The model is a structured diagram with three columns: SHAPE   NET (sketch)   SURFACE AREA FORMULA. Today they complete rows for: (1) Rectangular Prism (Cuboid), (2) Triangular Prism, (3) Square-Based Pyramid. Under each shape sketch, learners write one sentence connecting it to the Kisumu factory context — for example: 'A rectangular prism net</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Draw the three-column model template on the board before the phase begins. Give learners 6 minutes of silent individual work. Circulate and check that all three rows are complete and that sentences are specific (not just 'it is used in factories'). Prompt learners who write generic sentences: 'Which specific product at the Kisumu factory could this shape represent? How many cm<sup>2</sup> of metal does it need?' Cold-call 2 learners to read their contextual</p>                                                                                                                                                                                                                                                                                              | <p>Cumulative Model Building (Progressive Schematisation): The growing model functions as a personalised reference tool and a visible record of conceptual growth. Requiring a contextual sentence per shape prevents the model from becoming a passive formula list and reinforces the driving question connection. The 'UNRESOLVED' section at the bottom maintains productive tension and intellectual curiosity, ensuring learners see today's</p> | <p>Observable indicators: (1) All three model rows are complete with a sketch, correct formula, and a Kenya-relevant contextual sentence. (2) The 'UNRESOLVED' section is written in the learner's own words and names at least one shape from the phenomenon (cylinder, hemisphere, or frustum). (3) The thumbs survey gives the teacher a rapid whole-class confidence gauge — learners with 'thumbs down' should be identified for follow-up at the start of Lesson</p>                      |

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| <a href="#">Rectangular Prisms (Flexbook)</a><br>Source: CK-12<br><a href="#">Search ARES</a><br>for similar readings | is like a flat metal sheet cut at the factory that, when folded, forms a box-shaped water trough.' At the bottom of the model page, learners write: 'UNRESOLVED: We do not yet know the surface area of the cylindrical or frustum tanks in Kisumu — that is what we are investigating.' The teacher reminds the class: 'This model will grow every lesson. Do not throw it away — you will add to it in Lessons 2, 3, 4, and 5.' | sentences aloud. Close the lesson by returning to the original Predict Phase predictions still written on the board: 'Look at your prediction from the start of class. Has today's lesson changed your thinking? Thumbs up if yes, thumbs sideways if partly, thumbs down if your thinking has not changed.' Do a quick visual count and say: 'By the end of this unit, I predict all thumbs will be pointing firmly up — because we will have fully solved the Kisumu tank mystery.' | lesson as a starting point rather than a conclusion. | 2. |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners' predictions in Phase 1 reveal a clear distinction between surface area and volume, or did the majority conflate the two concepts? If conflation was widespread, plan a brief 5-minute clarification opener for Lesson 2 using a real Kenyan example — e.g., 'wrapping a mandazi box uses surface area; filling the box with mandazis uses volume.' (2) During the net activity (Phase 2), did any groups fail to include all faces when summing areas? Identify the specific face most commonly omitted (typically the base) and address this explicitly in the Lesson 2 warm-up. (3) Were learners able to derive the formulae independently through guided questioning, or did they require the teacher to provide the formula directly? If the latter, consider introducing a structured 'formula derivation scaffold' card for lower-attaining groups in Lesson 2. (4) Review the DQB sticky notes: Do the 'wonder' questions suggest learners are genuinely curious about cylinders and frustums (indicating readiness for Lesson 3), or are most questions still about prisms and pyramids (indicating the need for consolidation in Lesson 2)? (5) Did the Kenyan factory context feel authentic and engaging to learners, or did it feel contrived? Note 2–3 specific learner comments or questions that demonstrate genuine connection to the phenomenon for use in future lesson planning.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>What did I observe?</b>                   | Learners folded and unfolded pre-printed nets of a rectangular prism (a water trough), a triangular prism (a roof support beam), and a square-based pyramid (a decorative roof cap) — all modelled as flat metal sheets from the Kisumu factory. They measured and summed individual face areas to estimate total surface area before seeing any formula.                                               |
| <b>What did I learn?</b>                     | Learners derived and applied the surface area formulae for rectangular prisms ( $SA = 2(lw + lh + wh)$ ), triangular prisms ( $SA = bh + (a+b+c)l$ ), and square-based pyramids ( $SA = l^2 + 2ls$ ). They connected each algebraic term to a specific face of the net, and applied formulae to calculate the metal sheet needed for factory products.                                                  |
| <b>How does this explain the phenomenon?</b> | The total surface area of a prism or pyramid is the sum of the areas of all its faces, which can be found by unfolding the solid into a flat net. Different solid shapes — even if they enclose the same volume — can have very different surface areas. This is why the frustum tank in Kisumu uses less metal sheet than the cylinder-hemisphere tank despite holding the same 5,000 litres of water. |

## LESSON 2 (80 minutes): Surface Area of Cones and Spheres

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Purpose</b>              | To extend learners' understanding of surface area from flat-faced solids (Lesson 1) to curved solids — specifically cones and spheres — by deriving their surface area formulae and applying them to real Kenyan manufacturing and architectural contexts.                                                                                                                                                                                                                                                                                            |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) identify the components of the surface area of a cone (base circle and lateral/curved surface); (b) derive the formula for the lateral surface area and total surface area of a cone ( $SA = \pi r(r + l)$ , where $l$ is the slant height); (c) state and apply the formula for the surface area of a sphere ( $SA = 4\pi r^2$ ); (d) calculate the surface area of cones and spheres in real-world contexts.                                                                           |
| <b>Skills</b>               | Deriving geometric formulae from first principles; applying surface area formulae accurately; converting units; interpreting net diagrams of curved solids; problem-solving using cone and sphere surface area.                                                                                                                                                                                                                                                                                                                                       |
| <b>Attitudes</b>            | Curiosity about why curved surfaces require different formulae from flat-faced solids; appreciation of how geometry underpins cost-efficient manufacturing decisions by Kenyan engineers and fabricators.                                                                                                                                                                                                                                                                                                                                             |
| <b>Key Inquiry Question</b> | How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Purpose in Storyline</b> | In Lesson 1, learners computed the surface area of prisms and cylinders and began building their DQB around the anchoring phenomenon. In Lesson 2, they extend their evidence-gathering to curved solids — cones and spheres. By examining a conical roof turret found on Nairobi buildings and a spherical mandazi, learners derive the cone and sphere surface area formulae and add new evidence to their DQB, moving closer to explaining why the frustum tank at the Kisumu factory uses less metal than the composite cylinder-hemisphere tank. |
| <b>Safety Notes</b>         | No hazardous materials are used in this lesson. If learners are cutting paper nets, ensure scissors are handled safely and responsibly. Remind learners to handle any physical models (oranges, conical party hats) carefully to avoid injury.                                                                                                                                                                                                                                                                                                        |

### B. LESSON OVERVIEW

Learners begin by predicting how the surface area of a cone compares to that of a cylinder of the same height and base radius, connecting back to the Kisumu tank phenomenon. They then observe a physical/paper net of a cone to derive the lateral surface area formula ( $\pi rl$ ) and total surface area formula ( $\pi r(r + l)$ ), followed by a guided investigation using an orange peel to intuit the sphere formula ( $4\pi r^2$ ). In the Explain phase, learners solve structured problems set in Kenyan contexts — including calculating the metal sheeting needed for a conical roof turret in Nairobi and the material needed to coat a spherical mandazi mould. The DQB is updated with new evidence about curved surfaces. Learners close the lesson by revising their cumulative solid model to include cones and spheres, annotating it with the new formulae and connections to the tank phenomenon.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                             | Teacher Moves                                       | Sensemaking Strategy                                      | Formative Assessment Strategy                                |
|----------------------|----------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| <b>Predict Phase</b> | Learners are shown two images side by side on the board: (1) a | Display the two images prominently. Say: 'Before we | Prediction Tally (Activate-and-Confront): Recording class | Observable indicator: Every learner has written a prediction |

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| <p> <b>VIDEO:</b><br/> <a href="#">Surface area of a box (cuboid)</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p>                                            | <p>cylindrical water tank (as discussed in Lesson 1) and (2) a conical roof turret from a building in Nairobi's Westlands. Both have the same base radius (<math>r = 0.7</math> m) and the same vertical height (<math>h = 1.0</math> m). In pairs, learners write a prediction in their exercise books: 'Which solid do you think has the greater total surface area — the cylinder or the cone? Give a reason.' Learners also predict whether a sphere would have more or less surface area than both. After 3 minutes, pairs share predictions with the class. Teacher records class predictions as a tally on the board (Cone &gt; Cylinder / Cylinder &gt; Cone / Same) for later verification.</p>                                                                                                                                                                                                                                                                                                             | <p>calculate anything today, I want you to think like an engineer at a Nairobi construction firm. Look at these two shapes — same base, same height. Which one do you think needs more material to cover its surface? Write your prediction and a reason in your book. You have 3 minutes — go.' [Allow 3 minutes of silent individual think time.] After 2 minutes, say: 'Now share your prediction with your partner. Do you agree? You have 1 minute.' After sharing, cold-call 4 pairs (ensure gender balance): 'Pair 3 — what did you predict and why?' Record all predictions as a tally. Do NOT confirm or deny. Say: 'Hold onto your predictions — our investigation today will tell us who is right. Let us see what the mathematics says.'</p>                                                                                                                                                                              | <p>predictions publicly creates cognitive investment. When learners later discover the correct answer, the contrast between prediction and evidence deepens conceptual retention. This strategy also surfaces common misconceptions (e.g., 'taller means more surface area') that the teacher can address explicitly in the Explain phase.</p>                                                                                                                                                                                                             | <p>with a stated reason in their exercise book before discussion begins. Teacher circulates during the 3-minute window and scans books. Look for misconceptions such as 'cone has more area because it is pointy' or 'they are the same because same height and radius' — note these to address in Phase 3.</p>                                                                                                                                                                                                                                                                              |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volumes of cones intuition</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>PART A — CONE NET INVESTIGATION (30 min): Each group of 4 receives a pre-cut cone net (a circle and a sector of a larger circle) and a complete cone model. Groups are guided through the following steps: (1) Identify the two components of the cone's surface — the circular base (radius <math>r</math>) and the curved lateral surface (sector). (2) Measure the slant height <math>l</math> (the radius of the sector) using a ruler. (3) Notice that the arc length of the sector equals the circumference of the base circle (<math>2\pi r</math>). (4) Using the formula for the area of a sector (Area = <math>\frac{1}{2} \times \text{arc length} \times \text{radius}</math>), derive that Lateral Surface Area = <math>\frac{1}{2} \times 2\pi r \times l = \pi rl</math>. (5) Write the total surface area: SA = <math>\pi rl + \pi r^2 = \pi r(r + l)</math>. Groups record each step in a structured observation table provided on a worksheet. PART B — SPHERE INTUITION ACTIVITY (15 min):</p> | <p>Distribute cone nets and models. Say: 'Engineers and architects in Nairobi use cone-shaped turrets on buildings. Before they can order metal sheeting, they must know exactly how much curved surface that cone has. Your job right now is to figure out the formula from the net — not from memory.' Walk groups through Step 1: 'Lay your net flat. How many pieces do you see? What shape is each piece?' [WAIT 10 seconds.] Cold-call 2 groups. For Step 3, ask: 'What do you notice about the curved edge of the sector and the circle it must attach to?' [WAIT 15 seconds.] Cold-call 3 groups. If groups are stuck on the sector area link, prompt: 'We said arc length of the sector equals <math>2\pi r</math>. The sector area formula is <math>\frac{1}{2} \times \text{arc length} \times \text{radius}</math> of sector. What is the radius of the sector on your net?' For the orange activity, say: 'Peel your</p> | <p>Net-to-Formula Derivation (Guided Inquiry): Rather than stating formulas, learners reconstruct them from the physical net. This builds formula ownership and reduces dependence on rote memorisation. The orange peel activity uses a concrete, Kenya-familiar object (oranges are grown in Kisii and Meru) to make the abstract <math>4\pi r^2</math> formula tangible and memorable. Both strategies tie back to the tank phenomenon: understanding where formulae come from helps learners evaluate which tank shape is more material-efficient.</p> | <p>Observable indicator: Each group's observation worksheet shows a complete, correctly labelled derivation pathway — from sector area to <math>\pi rl</math> to <math>\pi r(r + l)</math>. Teacher checks 3–4 worksheets during the activity. Key checkpoint question (ask each group): 'Why is <math>l</math> the slant height and not the vertical height in the formula?' Correct answer: because <math>l</math> is the actual length along the curved surface of the cone, not the perpendicular height. Groups unable to articulate this receive a targeted prompt before Phase 3.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Each group receives an orange. They peel the orange carefully and try to flatten the peel onto four circles drawn on paper, each with the same radius as the orange's cross-section. Observation: the peel covers almost exactly 4 circles. Teacher confirms: Surface Area of sphere = $4\pi r^2$ . Learners record this in their notes alongside a labelled diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | orange carefully and try to lay the peel flat on the four circles. What do you notice?' [WAIT 10 seconds.] Cold-call 2 groups. Say: 'This is not a coincidence — Archimedes proved this over 2,000 years ago, and it holds for every sphere. Write the formula: $SA = 4\pi r^2$ .' Emphasise: 'No flat base — a sphere is all curved surface.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Explain Phase</b><br> <b>VIDEO:</b><br><a href="#">Surface area of a box (cuboid)</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Solid geometry word problems</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners apply their newly derived formulae to three structured problems set in Kenyan contexts, working individually then comparing with a partner: PROBLEM 1 (Cone): A craftsman in Nairobi is fabricating a decorative conical metal turret for a rooftop. The cone has base radius $r = 0.7$ m and slant height $l = 1.22$ m. Calculate the total surface area of the turret to determine how much metal sheet is needed. PROBLEM 2 (Sphere): A bakery in Kisumu uses a spherical mould to shape special occasion mandazi. The mould has a radius of 6 cm. Calculate the surface area of the mould to determine the amount of coating material required. PROBLEM 3 (Comparison — links to Lesson 1): Compare the total surface area of the Nairobi cone turret (Problem 1) with a cylinder of the same base radius ( $r = 0.7$ m) and vertical height ( $h = 1.0$ m) calculated in Lesson 1. Which requires less material? How does this connect to the Kisumu tank phenomenon? After individual work (12 min), a class discussion resolves Problem 3 and revisits the opening prediction tally. | Say: 'You now have the tools. Let us put them to work on real Kenyan problems.' Display all three problems on the board. Say: 'Work individually first — 12 minutes. Show every step: formula, substitution, calculation, units.' Circulate actively. For learners substituting vertical height instead of slant height in Problem 1, ask: 'Which length did you use? Is that the length along the surface of the cone or the height inside it?' [WAIT 10 seconds.] For Problem 3, after individual work, facilitate whole-class discussion: 'Who found that the cone has less surface area than the cylinder? Show us your numbers.' Cold-call 3 individuals to present their working. After Problem 3, return to the prediction tally on the board: 'Look at our predictions from the start of the lesson. What does the mathematics tell us now?' [WAIT 15 seconds.] Cold-call 2 learners to make the explicit link: 'So what does this mean for the frustum tank versus the cylinder-hemisphere tank in Kisumu?' | Worked Example to Real-World Transfer (Context-Anchored Practice): Each problem is deliberately embedded in a Kenyan context (Nairobi craftsman, Kisumu bakery) so that formula application feels purposeful rather than abstract. Problem 3 explicitly bridges Lesson 1 and Lesson 2 evidence, building the cumulative argument needed to explain the anchoring phenomenon. Returning to the prediction tally creates a structured 'confirm or revise' moment that consolidates conceptual change. | Observable indicators: (1) In Problem 1, learner correctly identifies $l = 1.22$ m as slant height and not $h = 1.0$ m, and obtains $SA \approx 4.22$ m <sup>2</sup> (Accept 4.20–4.24 m <sup>2</sup> ). (2) In Problem 2, learner applies $SA = 4\pi r^2$ with $r = 0.06$ m (or $r = 6$ cm consistently) and obtains $SA \approx 452.4$ cm <sup>2</sup> . (3) In Problem 3, learner makes an explicit written statement connecting the smaller surface area of the cone to the tank phenomenon. Teacher collects 6 exercise books (random sample) at the end of this phase for a quick review before the next lesson. |
| <b>Driving</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Each learner receives two sticky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Say: 'Our Driving Question asks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Driving Question Board —                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Observable indicator: Every                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| <p><b>Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area of a box (cuboid)</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Surface Area of Spheres (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>notes (or writes in their DQB section of their exercise book if physical notes are unavailable). On the first note (EVIDENCE colour): they write one piece of evidence gathered today — e.g., 'A cone with the same base and height as a cylinder has less total surface area.' On the second note (NEW QUESTION colour): they write one new question sparked by today's lesson — e.g., 'What about a shape that is part cone and part something else — like the frustum?' Learners post their notes on the class DQB (or share verbally). Teacher reads 3–4 evidence notes and 3–4 new questions aloud, explicitly linking the new questions to the frustum and composite tank shapes that will be investigated in upcoming lessons.</p>                                                                                                                                                                   | <p>how shape determines how much material is needed. Look at our DQB from Lesson 1. We added evidence about cylinders and prisms. What new evidence can we add today — about cones and spheres?' Distribute sticky notes (or direct learners to their DQB pages). Say: 'Two sticky notes — one EVIDENCE, one NEW QUESTION. You have 3 minutes.' After 3 minutes, collect and post notes. Read 3 evidence notes aloud and ask the class: 'Is this good evidence for answering our Driving Question? Why?' [WAIT 10 seconds each.] Read 3 new question notes. Say: 'I can see some of you are already asking about the frustum — the exact shape of the second tank in Kisumu. That question is coming very soon in our sequence. Keep it on the board — we will answer it.'</p>                            | <p>Evidence Accumulation (Iterative Sense-Making): The DQB makes the progression of the storyline visible. By physically adding evidence from each lesson, learners see their understanding growing toward a full explanation of the Kisumu phenomenon. New question notes ensure the DQB drives forward momentum rather than simply summarising past learning. This metacognitive practice develops the 'Learning to Learn' core competency.</p>                                                         | <p>learner has written at least one evidence statement that correctly references a formula or comparison from today's lesson (e.g., mentions <math>SA = \pi r(r + l)</math> or <math>SA = 4\pi r^2</math>, or a correct numerical comparison). Teacher scans notes during posting. If a learner's evidence note is vague or incorrect (e.g., 'cones are curved'), teacher asks a targeted clarifying question: 'What did you calculate today that proves cones use less material?'</p>                                          |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area of a box (cuboid)</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area of Spheres Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>  | <p>Learners return to the cumulative 'Surface Area Model' they began in Lesson 1 — a visual reference page in their exercise books (or a group poster) that maps all solids studied so far with their surface area formulae, diagrams, and Kenyan context connections. In Lesson 2, learners ADD to this model: (1) A labelled diagram of a cone showing <math>r</math>, <math>h</math>, and <math>l</math>, with the formula <math>SA = \pi r(r + l)</math> and an annotation linking <math>l</math> to the Nairobi turret context. (2) A labelled diagram of a sphere showing <math>r</math>, with the formula <math>SA = 4\pi r^2</math> and an annotation linking it to the Kisumu bakery mould context. (3) An updated 'Comparison Table' column showing that for equal base radius and height, Cone <math>SA &lt;</math> Cylinder <math>SA</math> — with a written sentence: 'This helps explain why</p> | <p>Say: 'Open your Surface Area Model from Lesson 1. Today we are adding two new solids — the cone and the sphere. Add labelled diagrams, formulae, Kenyan context, and update your comparison table. You have 8 minutes.' Circulate and look for: (a) correct labelling of slant height <math>l</math> vs vertical height <math>h</math> on the cone diagram; (b) no flat base drawn on the sphere diagram; (c) a correctly written comparison sentence. For early finishers: 'Can you sketch a frustum? It is like a cone with the top cut off. What parts would you need to find its surface area?' [WAIT 15 seconds before assisting.] In the final 2 minutes, cold-call 2 learners to share their updated model with the class. Close by saying: 'Your model is growing lesson by lesson. By the</p> | <p>Cumulative Model Revision (Progressive Knowledge Building): The running model page makes the growth of mathematical knowledge concrete and personal. Adding each new solid alongside its Kenyan context and a comparison note means learners are not just collecting formulae — they are building a coherent argument. The frustum sketch task for early finishers previews upcoming lessons and sustains intellectual momentum, directly advancing the storyline toward the anchoring phenomenon.</p> | <p>Observable indicator: The updated model page contains (1) a correctly labelled cone diagram with <math>l</math> clearly distinguished from <math>h</math>, (2) the correct formulae for both cone and sphere, (3) at least one Kenyan context annotation, and (4) an updated comparison statement. Teacher conducts a 'model gallery walk' for 2 minutes — examining 5–6 books — before dismissal. Any model missing the <math>l</math> vs <math>h</math> distinction is flagged for follow-up at the start of Lesson 3.</p> |

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|  | differently shaped tanks can have different surface areas even if designed for the same purpose.' Learners who finish early begin sketching what a frustum (truncated cone) might look like and hypothesise what its surface area formula might involve. | end of this unit, it will give you a complete toolkit to fully explain — and calculate — what is happening at that factory in Kisumu.' |  |  |
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the majority of learners correctly distinguish slant height ( $l$ ) from vertical height ( $h$ ) in the cone surface area formula? If not, plan a targeted 5-minute recap at the start of Lesson 3 using a physical cone model. (2) Were learners able to make an explicit verbal or written connection between the cone-vs-cylinder surface area comparison (Phase 3, Problem 3) and the Kisumu tank phenomenon? If the link was weak, revisit the phenomenon image at the start of Lesson 3 before proceeding. (3) How many new questions on the DQB mentioned the frustum or 'cut cone'? This signals readiness for Lesson 4 (frustum surface area). (4) Did the orange peel activity successfully build intuition for  $SA = 4\pi r^2$ ? If learners struggled with the connection, consider using a digital simulation (e.g., GeoGebra sphere net) in Lesson 2's revision or as a homework extension. (5) Review the 6 sampled exercise books: are learners showing correct unit usage ( $\text{cm}^2$ ,  $\text{m}^2$ )? Unit errors are a persistent source of calculation mistakes in KCSE examinations and should be addressed immediately.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| What did I observe?                                          | Learners unfolded a cone net and observed that the curved lateral surface is a sector of a circle. They measured the slant height and identified that the arc length of the sector equals the circumference of the cone's base. They also observed an orange peel laid flat over four circles of equal radius to the orange's cross-section.                                                                                                                                                                                                                                                    |
| What did I learn?                                            | The total surface area of a cone is $SA = \pi r(r + l)$ , where $r$ is the base radius and $l$ is the slant height (not the vertical height). The surface area of a sphere is $SA = 4\pi r^2$ , covering the entire curved surface with no flat face. For the same base radius and height, a cone has a smaller total surface area than a cylinder.                                                                                                                                                                                                                                             |
| How does this explain the phenomenon?                        | The cone surface area formula comes from adding the area of the circular base ( $\pi r^2$ ) to the lateral surface area derived from the sector ( $\pi rl$ ), giving $\pi r(r + l)$ . The sphere formula ( $4\pi r^2$ ) was derived intuitively from the orange peel activity and confirmed mathematically. These formulae explain why curved solids can enclose or cover space differently from flat-faced solids — a key piece of evidence for understanding why the Kisumu tank manufacturer can use different amounts of metal sheet for tanks of the same volume depending on their shape. |

## LESSON 3 (80 minutes): Surface Area of Frustums of a Cone

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | To derive and apply the formula for the surface area of a frustum of a cone, connecting it to the design of the frustum-shaped water tank in the anchoring phenomenon.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) identify a frustum as the solid formed when a cone is cut parallel to its base; (b) derive the surface area formula for a frustum of a cone using the slant heights of the original and removed cones; (c) calculate the total surface area of a frustum given its radii and slant height.                                                                                                                                                                                      |
| <b>Skills</b>               | By the end of the lesson, the learner should be able to: (a) apply the frustum surface area formula to solve real-world problems involving tanks, buckets, and containers; (b) interpret video-based evidence to extract mathematical relationships; (c) construct and annotate a diagram of a frustum showing all key dimensions.                                                                                                                                                                                                           |
| <b>Attitudes</b>            | The learner appreciates how geometric reasoning enables engineers and manufacturers in Kenya to make cost-effective decisions in designing storage solutions for rural communities.                                                                                                                                                                                                                                                                                                                                                          |
| <b>Key Inquiry Question</b> | How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | In Lesson 1, learners calculated the surface area of the composite cylinder-hemisphere tank. In Lesson 2, they explored how volume is conserved across different shapes. Now in Lesson 3, learners turn their attention to the frustum-shaped tank — the tank that puzzled them by using less metal. By deriving and applying the frustum surface area formula, learners gather the critical piece of evidence needed to explain why the frustum tank is cheaper to manufacture than the composite tank, even though both hold 5,000 litres. |
| <b>Safety Notes</b>         | No physical hazards. Ensure all learners can view the video clearly; adjust screen brightness and volume. Learners with visual impairments should be seated closest to the display. Handle rulers and compasses carefully during diagram construction.                                                                                                                                                                                                                                                                                       |

### B. LESSON OVERVIEW

Learners begin by predicting what happens to a cone's surface when it is cut parallel to its base, annotating a diagram of the Kisumu frustum-shaped tank. They then watch a curated YouTube video on the surface area of a frustum, pausing at key moments to extract the formula derivation. Using a Think-Pair-Share sensemaking strategy, learners derive the formula  $SA = \pi(R^2 + r^2) + \pi(R + r)l$  and apply it to calculate the surface area of the frustum water tank from the anchoring phenomenon. The lesson concludes with learners updating the Driving Question Board and revising their cumulative solid models to include the frustum, building a direct comparison with the composite tank from Lesson 1.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                 | Learner Experience                                                                                                             | Teacher Moves                                                                                                           | Sensemaking Strategy                                                                                                                    | Formative Assessment Strategy                                                                                         |
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| <b>Predict Phase</b><br> <a href="#">VIDEO: Volume of a sphere</a> | Learners receive a printed outline diagram of a frustum (labelled: 'Frustum-Shaped Tank, Kisumu Factory'). The diagram shows a | Distribute the frustum diagram handout. Say: 'Look at this diagram of the frustum-shaped tank from the Kisumu factory — | Prediction Annotation: Learners annotate a contextualised diagram before instruction, activating prior knowledge of cone geometry (from | Observable indicator: Each learner has labelled at least two dimensions on the frustum diagram and written a complete |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                         | <p>frustum with dotted lines indicating the removed top cone. Learners individually annotate the diagram, labelling what they believe are the key dimensions needed to find the surface area, and write a prediction: 'I think the formula for the surface area of this frustum will involve... because...'. Learners also write a one-sentence prediction answering: 'Do you think the frustum tank uses more or less metal sheet than the cylinder-hemisphere tank from Lesson 1? Why?' Predictions are written in exercise books before any discussion.</p>                                                                                                                                                                                                                                                                                                                   | <p>the tank that puzzled our Grade 10 visitors. Before we watch any video or do any calculation, I want you to think like an engineer. What measurements would YOU need to find how much metal sheet covers this tank?' Allow 10 seconds WAIT TIME. Say: 'Write your prediction in your book — label the diagram and write your sentence starting with: I think the formula will involve...'. Circulate silently for 4 minutes. Cold-call 3 learners: 'Amara, what did you label?' / 'Odhiambo, what measurements did you choose and why?' / 'Wanjiku, do you agree with the prediction we just heard — why or why not?' Record key prediction terms on the board (e.g., radius, slant height, height). Do NOT confirm or correct — say: 'Excellent thinking. Let's gather evidence to test these predictions.'</p> | <p>Lesson 2) and creating cognitive dissonance. This strategy surfaces misconceptions (e.g., confusing vertical height with slant height) and gives learners a personal stake in the video evidence they are about to gather.</p>                                                                                                                                                                                                                  | <p>prediction sentence. Teacher scans books during circulation — look for learners who label only height (missing slant height or radii) as a flag for targeted questioning during the video phase.</p>                                                                                                                                                                                                                |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Volume of triangular prism &amp; cube</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners watch a curated YouTube video on the surface area of a frustum of a cone (approx. 8–10 minutes). The video is paused three times at teacher-designated moments: (1) after the frustum is introduced — learners sketch and label the frustum in their books; (2) after the derivation of the curved surface area formula <math>\pi l(R + r)</math> — learners write the formula and compare with their prediction; (3) after the complete formula <math>SA = \pi(R^2 + r^2) + \pi l(R + r)</math> is stated — learners re-examine their diagram and label slant height <math>l</math>, large radius <math>R</math>, and small radius <math>r</math>. A structured observation guide is projected alongside the video with three sentence stems: 'A frustum is formed by...', 'The slant height <math>l</math> is found using...', 'The total surface area formula</p> | <p>Before starting the video say: 'As you watch, fill in the three sentence stems projected on the board. Pause your thinking at each stop — I will freeze the video and ask you to write before we move on.' Play video to Pause Point 1 (~2 min). Freeze. Say: 'Sketch what you see. Label the two circular faces — what are their radii called?' WAIT TIME 10 seconds. Cold-call 2 learners: 'Korir, what did you sketch and label?' / 'Fatuma, how does your diagram compare to your prediction from Phase 1?' Resume. Pause at Point 2 (~5 min). Freeze. Say: 'Write the curved surface area formula — <math>\pi l(R + r)</math> — in your book. Where have you seen something like this before? Think about the cone formula from our earlier work.'</p>                                                      | <p>Structured Note-Taking with Sentence Stems: By providing three sentence stems aligned to the video's key conceptual moments, learners are guided to process the derivation actively rather than passively watching. The three pause points create retrieval practice opportunities and directly test learner predictions, building the habit of comparing prior conceptions against new evidence — a core science and engineering practice.</p> | <p>Observable indicator: At each pause point, learners have written responses in their books. Teacher scans during pauses — specifically check that learners have correctly identified <math>R</math> (large base radius), <math>r</math> (small top radius), and <math>l</math> (slant height of the frustum, NOT the original cone). Learners who write only one radius are flagged for Pair support in Phase 3.</p> |



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| <p><b>Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of a sphere</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p>                    | <p>note (or card): one piece of NEW EVIDENCE gathered today and one NEW QUESTION that arose from today's lesson. Evidence notes are placed in the 'What We Now Know' column; question notes in the 'What We Still Wonder' column. As a class, learners review the board and draw arrows connecting today's evidence (frustum surface area formula) to the driving question: 'How does the shape of a solid determine how much material is needed to build it?'</p>                                                                                                                                                                                                                                        | <p>tank. In Lesson 2 we posted evidence about volume. What new evidence can we post today?' WAIT TIME 10 seconds. Distribute sticky notes. Say: 'Write ONE piece of evidence — something you can now calculate that you could not before — and ONE new question your work today raised. You have 3 minutes.' After writing: invite 4 learners (cold-call) to read their evidence note aloud before posting it. Ask the class: 'Where should this go on the board? Does it help us answer the driving question directly, or does it raise a new sub-question?' Use a red marker to draw arrows from today's evidence notes to the main driving question card. Say: 'We are building a case. Every lesson adds another brick. What shape are we still missing from our investigation?' (Prompt learners to anticipate upcoming lessons on spheres or composite solids.)</p> | <p>physically placing evidence and questions on the board, learners externalise their thinking and see how each lesson contributes to the overarching investigation. Drawing arrows models scientific reasoning — linking evidence to claims — and helps learners see the frustum lesson not as isolated content but as a chapter in a longer evidentiary story.</p>                                                                                                  | <p>note. Teacher reads notes during posting to identify misconceptions (e.g., a learner who writes 'frustums have more surface area than cylinders' as a general rule — this overgeneralisation is addressed verbally: 'Interesting — is that always true, or only for these specific dimensions?'). The quality of new questions also reveals depth of understanding.</p>                                                                                                                             |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Volume of a sphere</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners retrieve their cumulative 'Solid Shape Model' — a growing annotated diagram begun in Lesson 1 that now shows the composite cylinder-hemisphere tank. Today, learners add a side-by-side diagram of the frustum tank, labelled with R, r, l, and the surface area formula. Beneath both diagrams, learners write a 'Designer's Note' in the voice of a Kisumu tank manufacturer: 'I would choose the frustum shape because... / I would choose the composite shape because...', justifying their choice using the surface area values calculated in Phase 3. Learners who finish early are challenged to sketch a third tank shape (e.g., a pure cylinder) and estimate which of the three</p> | <p>Say: 'Open your cumulative model from Lesson 1. Today we are adding the frustum to our model. Draw the frustum side-by-side with the composite tank and label all key dimensions.' Allow 4 minutes for drawing. Circulate and check that learners label R (large radius), r (small radius), and l (slant height) — NOT vertical height h. After 4 minutes: Say: 'Now write your Designer's Note. You are the factory manager in Kisumu. You have the data — the surface areas, the costs. Which shape do you manufacture and why?' WAIT TIME 12 seconds. Cold-call 2 learners to read their Designer's Note: 'Achieng, what would you manufacture and what is your evidence?' / 'Njoroge, do you</p>                                                                                                                                                                   | <p>Cumulative Model Revision with Role-Play Justification: Adding to a running annotated model helps learners see knowledge as cumulative and connected rather than lesson-by-lesson fragments. The Designer's Note strategy uses disciplinary role-play to require learners to argue from evidence — a key NGSS practice ('Engaging in Argument from Evidence') — and grounds the mathematics in the real-world decision-making context of Kenyan manufacturing.</p> | <p>Observable indicator: The frustum diagram in learner books is correctly labelled with R, r, and l (not h), and the surface area formula is written beneath it. The Designer's Note contains at least one specific numerical value (e.g., 'the frustum uses 7.03 m<sup>2</sup> of metal, saving Ksh X compared to the composite tank') as evidence for the choice. Learners who write only qualitative justifications ('it looks smaller') are prompted: 'Can you add a number to support that?'</p> |

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|  | shapes would use the least material for the same volume. | agree? Would any situation change your decision?' Close the lesson by saying: 'Next lesson, we will investigate the volume of the frustum — then we will have ALL the evidence we need to fully answer our driving question.' |  |  |
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### D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were learners able to distinguish between the slant height of the frustum ( $l$ ) and the vertical height ( $h$ )? If many confused these, consider adding a dedicated 3-minute mini-activity at the start of Lesson 4 where learners use Pythagoras' theorem to derive  $l$  from  $h$ ,  $R$ , and  $r$ . (2) Did the video pause points generate genuine engagement, or did learners copy passively? If passive copying was observed, restructure Phase 2 for future delivery by assigning different sentence stems to different table groups, creating a jigsaw-style knowledge share. (3) Were learners able to write a meaningful comparison sentence connecting frustum surface area to the composite tank from Lesson 1? If learners could not recall the Lesson 1 value, consider keeping a running 'Evidence Wall' poster in the classroom that displays key calculated values from each lesson for reference. (4) Did the cost calculation (Ksh 850/m<sup>2</sup>) generate authentic engagement with the Kenya context? If learners seemed disengaged, invite a local hardware or construction materials supplier price list for the next lesson to ground the numbers further. (5) Review the DQB — do the new questions posted suggest learners are ready to investigate frustum volume in Lesson 4, or do they need more consolidation on surface area first?

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>What did I observe?</b>                                   | Learners watched a video showing how a frustum is formed by cutting a cone parallel to its base, and observed the step-by-step derivation of the curved surface area formula $\pi l(R + r)$ and the total surface area formula $SA = \pi(R^2 + r^2) + \pi(R + r)l$ , pausing at three key moments to sketch, label, and record the formula.                                                                    |
| <b>What did I learn?</b>                                     | Learners learned that a frustum of a cone has two circular faces (radii $R$ and $r$ ) and a curved lateral surface, and that its total surface area is calculated using $SA = \pi(R^2 + r^2) + \pi(R + r)l$ ; they applied this formula to calculate that the Kisumu frustum-shaped water tank has a total surface area of approximately 7.03 m <sup>2</sup> , costing approximately Ksh 5,975 in metal sheet. |
| <b>How does this explain the phenomenon?</b>                 | Learners explained why the frustum-shaped tank in the anchoring phenomenon uses less metal sheet than the composite cylinder-hemisphere tank by comparing their calculated surface areas, and justified — in the role of a Kisumu tank manufacturer — which shape they would choose to produce based on cost and material efficiency evidence.                                                                 |

## LESSON 4 (80 minutes): Surface Area of Frustums of a Pyramid

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | To enable learners to derive and apply the formula for the surface area of a frustum of a pyramid in real-world manufacturing and storage contexts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to derive the formula for the surface area of a frustum of a pyramid and apply it to solve problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Skills</b>               | By the end of the lesson, the learner should be able to calculate the surface area of a frustum of a pyramid accurately, connecting geometric reasoning to practical material-estimation problems.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Attitudes</b>            | By the end of the lesson, the learner should appreciate how mathematical derivation of surface area formulas empowers engineers and manufacturers in Kenya to design cost-effective storage structures.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Key Inquiry Question</b> | How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Purpose in Storyline</b> | In Lesson 3, learners explored the surface area of cones and frustums of cones. In this lesson, learners extend their understanding to frustums of pyramids — the shape used in traditional grain stores in Kitale and in some locally fabricated metal storage hoppers. By deriving the formula for the lateral and total surface area of a frustum of a pyramid and applying it to contextualised problems, learners build a richer picture of how a solid's shape (not just its volume) determines the amount of material needed to construct it — directly advancing their investigation of the Kisumu tank manufacturer's puzzle. |
| <b>Safety Notes</b>         | No physical hazards anticipated. Ensure rulers and compasses are used carefully when constructing net diagrams. Supervise safe use of digital devices when streaming the YouTube video.                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### B. LESSON OVERVIEW

Learners begin by briefly recalling the frustum-of-cone surface area formula from Lesson 3 and connecting it to the anchoring phenomenon. They then watch a focused YouTube video on the surface area of a frustum of a pyramid, pausing at key moments to annotate diagrams. Working in groups, they derive the formula by 'unfolding' a frustum of a pyramid into its net — a base rectangle/square, a top rectangle/square, and four trapezoidal lateral faces — and generalising to a formula. They apply the formula to a Kenyan context problem (metal sheet needed for a frustum-shaped grain store in Kitale), compare their answers with peers, and update the Driving Question Board with new evidence about the relationship between shape and surface area.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                   | Learner Experience                                                                                                                                                      | Teacher Moves                                                                                                                                          | Sensemaking Strategy                                                                                                                                   | Formative Assessment Strategy                                                                                                                                        |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Surface area using a net: triangular prism</a> | Learners are shown a large image of two structures side by side: (A) a cylindrical grain store with a conical roof in Siaya County, and (B) a frustum-of-pyramid-shaped | Display the image on the board/projector. Say: 'Look carefully at these two grain stores. Both hold the same volume. In your exercise book, write your | Think-Pair-Share — Activates prior knowledge of surface area concepts from Lessons 1–3, surfaces learner misconceptions about the relationship between | Observable indicator: Learners' written predictions reference at least one geometric property (e.g., number of faces, slant height, shape of lateral faces). Teacher |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Pyramids Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                   | <p>metallic grain hopper used in Kitale. Both are labelled as holding the same volume of grain. Each learner silently writes a prediction in their exercise book answering: 'Which structure do you think uses more sheet metal to build — and why?' They then share predictions with a shoulder partner (Think-Pair-Share). Selected pairs share with the class; the teacher records key prediction words on the board without affirming or correcting.</p>                                                                                                                                                                                                                                                                                                 | <p>prediction — which one uses more metal sheet, and give a mathematical reason.' Allow 10 seconds of silent thinking (WAIT TIME). After writing time (~2 minutes), say: 'Turn to your shoulder partner and share your prediction.' After 90 seconds, cold-call 3 pairs (vary by gender and seating zone). Record key words — e.g., 'more faces', 'slanted sides', 'trapezoidal panels' — on the board under the heading 'Our Predictions'. Say: 'We will return to these predictions at the end of the lesson to see who was right and why.'</p>                                                                                                                                                                                                                                                                                                                                                                                          | <p>volume and surface area, and creates cognitive dissonance that motivates the derivation activity to follow.</p>                                                                                                                                                                                               | <p>circulates and notes learners who write only volume-based reasoning (e.g., 'same volume means same surface area') as these learners need targeted questioning during the Explain phase.</p>                                                                                                                                                                                           |
| <p><b>Observe Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Surface area word problem example</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>HTML:</b><br/> <a href="#">Surface Area and Volume of Pyramids (PLIX)</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners watch a curated YouTube video on 'Surface Area of a Frustum of a Pyramid' (approx. 10–12 minutes). Before viewing, each learner receives a structured observation worksheet with three columns: (1) What I notice about the shape/net, (2) Key measurements labelled in the video, (3) Questions I have. The teacher pauses the video at three predetermined moments: (a) when the net of the frustum of a pyramid is first shown, (b) when the formula for lateral face (trapezium) area is introduced, and (c) when the full surface area formula is assembled. At each pause, learners annotate their worksheet silently for 30 seconds before the video resumes. After the video, learners compare worksheet notes with a group partner.</p> | <p>Before playing the video, say: 'As you watch, complete your three-column worksheet. Do not copy every word — focus on the shape of the net, the labels used, and any steps you find surprising or unclear.' Play the video. Pause at moment (a) and say: 'Pause here — look at the net. What shapes do you see? Write them in column 1. You have 30 seconds.' (WAIT TIME: 30 seconds.) Resume. Pause at moment (b) and say: 'The video just showed how the area of one lateral face is calculated. Write the formula for the area of a trapezium in column 2 — you have 30 seconds.' (WAIT TIME: 30 seconds.) Pause at moment (c) and say: 'The full formula is about to be summarised. Before it appears — can you predict it from your notes? Write your prediction.' After the video, cold-call 2 learners to read out one observation each. Say: 'Keep these worksheets — you will need them for the derivation activity next.'</p> | <p>Structured Note-Taking with Planned Pauses — Rather than passive viewing, the three-pause protocol forces learners to process and record information in real time. Connecting the net diagram to known shapes (squares, rectangles, trapeziums) bridges from prior knowledge to new formula construction.</p> | <p>Observable indicator: Learners correctly identify the net of a frustum of a pyramid as comprising two rectangular/square faces and four trapezoidal lateral faces on their worksheet. Teacher scans worksheets during the partner-comparison step; learners who have blank or incorrect net descriptions are grouped with stronger peers during the Explain phase group activity.</p> |

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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area using a net: triangular prism</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface area word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners work in groups of four to derive the surface area formula of a frustum of a pyramid from first principles using a guided-discovery worksheet. The worksheet provides: a labelled diagram of a frustum of a square pyramid (bottom side length <math>a</math>, top side length <math>b</math>, slant height <math>l</math> of each trapezoidal face, vertical height <math>h</math>), and the following scaffolded steps: (1) Identify and name each face of the frustum. (2) Write the area formula for the bottom face. (3) Write the area formula for the top face. (4) Derive the area of one trapezoidal lateral face using Area = <math>\frac{1}{2}(a + b) \times l</math>. (5) Write the total lateral surface area for all four faces. (6) Assemble the complete surface area formula: <math>SA = a^2 + b^2 + 4 \times \frac{1}{2}(a + b) \times l = a^2 + b^2 + 2l(a + b)</math>. Groups then solve the Kitale contextualised problem: 'A frustum-shaped grain store in Kitale has a square base of side 2.4 m, a square top of side 1.2 m, and a slant height of 1.5 m on each triangular face. Calculate the total sheet metal (in <math>m^2</math>) needed to construct the store, including the base but excluding the top opening.' Groups present their derivation steps on mini whiteboards or manila paper for a gallery walk.</p> | <p>Distribute the guided-discovery worksheet. Say: 'In your groups, you are going to derive — not just memorise — the surface area formula for a frustum of a pyramid. Use your observation worksheet from the video as a reference.' Circulate and monitor step 4; if groups struggle with the trapezium area, prompt: 'What is the formula for the area of a trapezium? Which two lengths are the parallel sides here?' (WAIT TIME: 15 seconds before assisting.) After 15 minutes of group work, say: 'Write your final formula on your manila paper and stick it up.' Conduct a 5-minute gallery walk. Cold-call 3 groups (different from those called in Phase 1) to explain their derivation step-by-step. Ask probing questions: 'Why do we multiply by 4?' and 'What changes in this formula if the base is a rectangle, not a square?' After the gallery walk, display the standard formula on the board: <math>SA = a^2 + b^2 + 2l(a + b)</math> [square base] and generalise for rectangular base: <math>SA = 2(lw) + 2(lw) + \text{lateral faces}</math>. Bring the class back to the anchoring phenomenon: 'Think about the frustum tank in Kisumu. The manufacturer uses a frustum of a pyramid for metal sheet panels — does our new formula help explain why the frustum tank might use less metal than the cylinder-hemisphere tank?'</p> | <p>Guided Discovery with Gallery Walk — Learners construct the formula themselves rather than receiving it, deepening procedural and conceptual understanding. The gallery walk creates peer accountability and exposes learners to multiple derivation approaches, reinforcing that there is a logical path from geometric properties to algebraic formula.</p> | <p>Observable indicator: Each group's manila paper shows a correctly assembled surface area formula with all terms labelled, and the Kitale problem answer is within <math>\pm 0.1 m^2</math> of the correct value (<math>SA = 2.4^2 + 4 \times \frac{1}{2}(2.4 + 1.2) \times 1.5 = 5.76 + 10.8 = 16.56 m^2</math>, minus the top opening = <math>5.76 + 4 \times \frac{1}{2}(2.4+1.2) \times 1.5 = 5.76 + 10.8 = 16.56 m^2</math> total including base). Teacher notes groups that skip the top-face exclusion as a common error to address in the next lesson.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area using a net: triangular prism</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Each learner receives two sticky notes (or writes in a designated section of their exercise book if sticky notes are unavailable). On the first sticky note (green), they write one piece of new evidence they now have that helps answer the driving question: 'How does the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Hand out sticky notes. Say: 'Take 3 minutes — silently. Green note: what new evidence do you have today that helps answer our big driving question? Orange note: what new question does today's lesson raise for you?' (WAIT TIME: 3 minutes silent writing.) Ask</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Driving Question Board Update — This metacognitive strategy makes learning progress visible and cumulative. By separating 'evidence' from 'new questions', learners practise distinguishing between what they know and what they still need to investigate, a</p>                                                                                             | <p>Observable indicator: Learners' green sticky notes reference a specific mathematical relationship (e.g., 'the formula <math>SA = a^2 + b^2 + 2l(a+b)</math> shows that slant height and base size both affect surface area independently of volume'). Vague responses such as 'I learned about</p>                                                                                                                                                                                                                                                                |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                               | <p>shape of a solid determine how much material is needed to build it and how much it can hold?' On the second sticky note (orange), they write one new question that today's lesson raised for them. Learners post their notes on the class Driving Question Board under the columns 'New Evidence' and 'New Questions'. The teacher reads 3–4 contributions aloud and connects them explicitly to the lesson sequence: 'In Lesson 3 we found the surface area of a frustum of a cone. Today we added the frustum of a pyramid. We are building a toolkit of formulas. What solid shapes are still missing from our toolkit?'</p>                                                                                                                                                                                                                      | <p>learners to post notes on the DQB. Read aloud 3 green and 2 orange contributions. For each green contribution, ask the class: 'Do you agree this is evidence? Give a thumbs up or thumbs down.' For orange questions, say: 'Which of these questions do you think we will answer in the next few lessons?' Accept 2–3 student responses. Close by pointing to the anchoring phenomenon image: 'Remember our Kisumu factory. The students visiting the factory noticed the frustum metal tank uses less sheet metal. We now have mathematical tools — for both frustum of cone AND frustum of pyramid — to start calculating and comparing. Next lesson, we investigate the composite cylinder-hemisphere tank.'</p>                                                                                                                      | <p>core science and mathematical reasoning practice.</p>                                                                                                                                                                                                                                                                                                                                                            | <p>frustums' indicate the learner needs re-engagement with the phenomenon connection during the next lesson's warm-up.</p>                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area using a net: triangular prism</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Pyramids Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to a cumulative 'Shape-Surface Area-Volume Model' diagram that they began in Lesson 1 — a large annotated table/diagram in their exercise books that tracks, for each solid studied: the solid name, a sketch of its net, the surface area formula, and a Kenyan real-world example. Today, learners add a new row for 'Frustum of a Pyramid', sketching the net, recording the derived formula <math>SA = a^2 + b^2 + 2l(a + b)</math> (for square base), and noting the Kitale grain store as the real-world example. They then write a two-sentence 'model statement' connecting today's learning to the anchoring phenomenon: e.g., 'A frustum of a pyramid has a surface area that depends on both the top and bottom base lengths and the slant height of its trapezoidal faces. This means the manufacturer in Kisumu can</p> | <p>Say: 'Open your cumulative model from Lesson 1. We are adding today's solid. Fill in the row for frustum of a pyramid — net sketch, formula, and Kitale grain store as the example. You have 4 minutes.' (WAIT TIME: 4 minutes.) Then say: 'Now write your two-sentence model statement linking today's learning to the Kisumu factory question. Be specific — use the words slant height, base length, and surface area.' After 2 minutes of writing, say: 'Read your statement to your partner. Partner: does the statement use at least two mathematical terms and connect to the Kisumu factory? Give one piece of feedback.' Cold-call 2 learners to read their revised model statements to the class. Close: 'Our model is growing lesson by lesson. We can now compare surface areas of cones, frustums of cones, frustums of</p> | <p>Cumulative Model Revision — The progressive model diagram makes conceptual growth tangible and forces learners to connect each new formula to both the net geometry and a real Kenyan context. The two-sentence model statement requires learners to synthesise procedural knowledge (formula) with conceptual understanding (what variables affect surface area), directly addressing the driving question.</p> | <p>Observable indicator: Learners' cumulative model row for 'Frustum of a Pyramid' contains a correctly labelled net sketch (square/rectangle top, square/rectangle base, four trapeziums), the correct formula, and a Kenyan real-world example. The two-sentence model statement explicitly names at least two variables (slant height and base length) and references the anchoring phenomenon. Learners who omit the net sketch or who copy the cone formula incorrectly are identified for targeted revision in Lesson 5's warm-up.</p> |

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|  | reduce sheet metal usage by adjusting these dimensions independently of the tank's volume.' Learners share their model statements with a partner for peer feedback before a final individual revision. | pyramids, and more. In Lesson 5, we will bring in the composite solids — and that is when we can truly start to answer our big question about the Kisumu tanks.' |  |  |
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| D. TEACHER REFLECTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>After the lesson, reflect on the following questions: (1) During the guided-discovery group work, which steps in the derivation caused the most confusion — was it identifying the trapezoidal faces, applying the trapezium area formula, or assembling the full expression? What targeted scaffolding will you provide in Lesson 5's warm-up? (2) Review the green sticky notes from the DQB update: do learners' 'evidence statements' show genuine understanding that surface area and volume are independent properties of a solid? If most statements are procedural ('I can use the formula') rather than conceptual ('slant height and base size affect surface area independently of volume'), plan a brief conceptual discussion for the start of Lesson 5. (3) Did all learner groups correctly exclude the top opening in the Kitale grain store problem? If not, this is a persistent real-world application error — consider a short 'what faces do we include?' checklist tool for future lessons. (4) Were there learners who struggled to connect the frustum-of-pyramid formula to the Kisumu anchoring phenomenon during model building? Note their names and ensure they are cold-called early in Lesson 5 to rebuild that connection.</p> |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>What did I observe?</b>                                   | Learners watched a YouTube video on the surface area of a frustum of a pyramid, pausing to annotate a three-column observation worksheet identifying the net components (two square/rectangular faces and four trapezoidal lateral faces) and the steps used to assemble the surface area formula.                                                                                   |
| <b>What did I learn?</b>                                     | Learners derived the surface area formula for a frustum of a square pyramid — $SA = a^2 + b^2 + 2l(a + b)$ — by decomposing the solid into its net, calculating the area of each face type, and summing them. They applied this formula to calculate the sheet metal required for a frustum-shaped grain store in Kitale.                                                            |
| <b>How does this explain the phenomenon?</b>                 | Learners explained, using their derived formula and their cumulative solid model, how the slant height and the top and bottom base lengths of a frustum of a pyramid independently determine its surface area — regardless of the volume enclosed — connecting this insight to the anchoring phenomenon of the Kisumu tank manufacturer's choice of shape to minimise material cost. |

## LESSON 5 (80 minutes): Surface Area of Composite Solids

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | To extend learners' understanding of surface area by computing the total exposed surface of composite solids — solids formed by combining two or more basic 3D shapes — and to connect this skill directly to understanding why the cylinder-hemisphere water tank from the Kisumu factory requires more metal sheet than the frustum tank, despite holding the same volume.                                                                                                                                                                                                                               |
| <b>Knowledge</b>            | Learners will know that: (a) a composite solid is formed by joining two or more basic solids; (b) when two solids are joined, the faces at the junction are hidden and must be subtracted from the total surface area; (c) the surface area of a composite solid = sum of surface areas of all component solids – 2 × area of each hidden (joined) face; (d) the cylinder-hemisphere tank's surface area can be computed systematically by identifying each exposed region.                                                                                                                                |
| <b>Skills</b>               | By the end of the lesson, learners should be able to: (a) identify the component solids in a composite solid and label each exposed surface region; (b) determine which faces are hidden at junctions and subtract their areas correctly; (c) calculate the total surface area of composite solids including the cylinder-hemisphere tank; (d) compare the surface area of the cylinder-hemisphere tank with that of the frustum tank (from Lessons 3–4) holding the same volume.                                                                                                                          |
| <b>Attitudes</b>            | Learners will: (a) appreciate that systematic mathematical decomposition of complex shapes leads to accurate and efficient solutions; (b) recognise that the choices engineers in Kenya make about tank shape are grounded in precise mathematical reasoning; (c) show persistence and care when managing multi-step surface area calculations.                                                                                                                                                                                                                                                            |
| <b>Key Inquiry Question</b> | How does the shape of a solid affect its surface area and volume? / How do we apply surface area and volume of solids in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | In Lessons 3 and 4, learners calculated the surface area and volume of the frustum-shaped tank. In Lesson 5, they now turn their attention to the other tank at the Kisumu factory — the composite cylinder-hemisphere tank. By computing its surface area and comparing it with the frustum's surface area (same volume, 5,000 L), learners gather critical evidence to answer why the factory's frustum tank uses noticeably less metal sheet. This lesson is the pivot point in the evidence-gathering sequence: learners for the first time have surface area figures for BOTH tanks in front of them. |
| <b>Safety Notes</b>         | No hazardous materials are used in this lesson. Ensure shared calculators and digital devices are handled responsibly. If the YouTube video is played, ensure screen brightness is comfortable and volume is set appropriately for the classroom environment. Learners with visual impairments should be seated near the display screen.                                                                                                                                                                                                                                                                   |

### B. LESSON OVERVIEW

Lesson 5 of 10 in the evidence-gathering sequence for the anchoring phenomenon (Kisumu water tank factory). Learners open by predicting whether the cylinder-hemisphere composite tank will have a larger or smaller surface area than the frustum tank they computed in Lessons 3–4. They then watch a focused YouTube video segment on the systematic strategy for finding the surface area of composite solids (identify components → subtract hidden faces → sum exposed surfaces). Using this strategy, they work through two applications: (1) a Kericho tea canister (cylinder + cone lid) as a scaffolded practice context, and (2) the cylinder-hemisphere water tank from the anchoring phenomenon, using the same volume-equivalent dimensions established in previous lessons. Learners record both surface area results and compare them directly to the frustum surface area, updating the Driving Question Board with new evidence. The lesson closes with a cumulative model revision in which learners annotate a diagram of both tanks showing where material is 'saved' or 'added' by shape choice.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                   | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Cylinder volume &amp; surface area</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Volume and surface area of cylinders</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners are shown a side-by-side diagram of the two Kisumu factory tanks: (A) the cylinder-hemisphere composite tank and (B) the frustum tank. Both are labelled as holding 5,000 litres. Learners individually write a prediction in their exercise books: 'Which tank do I think has a larger surface area, and why?' They then share their prediction with a shoulder partner using the sentence frame: 'I predict tank ____ has a larger surface area because ____.' Selected pairs share aloud. Predictions are NOT confirmed yet — they are posted on a class Predict Wall (sticky notes or chalkboard column). | Display the two-tank diagram on the board (drawn or projected). Say: 'Before we calculate anything today, I want your best mathematical guess. Look at both tanks. Both hold exactly 5,000 litres — same volume. Which one do you think needs more metal sheet to build? Write your prediction and your reasoning in your book. You have 2 minutes — go.' [Allow 2 minutes silent individual thinking.] After writing, say: 'Share with your shoulder partner using this frame: I predict tank ____ has a larger surface area because ____.' [Allow 90 seconds.] Cold-call 4 pairs — 2 who predict cylinder-hemisphere is larger, 2 who predict frustum is larger. Record predictions on the board under 'Predict Wall' columns. Say: 'We are going to test every single one of these predictions with mathematics today. Let's gather our evidence.' | Predict-then-Investigate (Anchored Prediction): By committing to a written prediction tied to the anchoring phenomenon before instruction, learners activate prior geometric intuition about surface area vs. volume. This creates productive cognitive dissonance that the lesson's evidence will resolve, deepening retention of the composite surface area concept. | Observable indicator: Each learner has a written prediction with a geometric or mathematical justification (not just a guess). Teacher scans books during pair-share. Red flag: learners who cannot articulate any reason — teacher prompts: 'Think about which tank has more curved surfaces or flat faces exposed.'                                                                                                                                    |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Cylinder volume &amp; surface area</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Surface Area and Volume of Cylinders Practice</a><br>Source: CK-12<br> <a href="#">Search ARES</a>                                      | Learners watch a targeted YouTube video segment (recommended: a composite solids surface area tutorial, e.g., from Math with Mr. J or similar, ~8–10 minutes) that demonstrates the three-step strategy: Step 1 — Identify and list all component solids. Step 2 — Identify hidden/joined faces and calculate their areas. Step 3 — Sum all exposed surface areas. During the video, learners complete a structured Video Observation Guide with three columns: Component Solid   Formula Used   Hidden Face? (Yes/No). After the                                                                                      | Before playing the video, distribute the Video Observation Guide. Say: 'As you watch, fill in each component solid you see, the formula the presenter uses, and whether that face is hidden at a joint. You will use these notes for your calculations, so be accurate.' Play the video (8–10 minutes). Pause at the moment the presenter subtracts the hidden circular face and say: 'Pause here — why are we subtracting this area? Discuss with your neighbour for 30 seconds.' [Allow 30 seconds.] Restart video. After the video, distribute or display the                                                                                                                                                                                                                                                                                      | Structured Video Observation + Diagram Annotation: The Video Observation Guide transforms passive video watching into active evidence collection. Diagram annotation of the Kericho canister then bridges from the abstract video example to a Kenya-relevant context, ensuring learners can transfer the strategy before applying it to the phenomenon tanks.         | Observable indicator: Learners' canister diagrams correctly identify the shared circular face ( $r$ = radius of cylinder = radius of cone base) as the ONE hidden face, and all other faces as exposed. Teacher checks group diagrams during the 4-minute activity. Red flag: groups that colour the cylinder's flat top as 'exposed' — teacher asks: 'If I glue the cone base onto the cylinder top, can I still see or access that face from outside?' |

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| <a href="#">for similar readings</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>video, learners examine a physical or drawn model of the Kericho tea canister (a cylinder with a shallow cone lid — a familiar composite solid from Kenyan household contexts). In groups of 3, they label a blank diagram of the canister: cylinder lateral surface, cylinder top (hidden — joined to cone base), cone lateral surface (slant face), cylinder base (open or closed depending on design). They are NOT yet calculating — only identifying and labelling surfaces.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Kericho tea canister diagram. Say: 'In your groups of three, label every surface of this canister. Use a different colour or code for: exposed surfaces you INCLUDE and hidden surfaces you SUBTRACT. You have 4 minutes.' [Allow 4 minutes.] Cold-call 2 groups to share their labelling on the board. Correct misconceptions immediately, especially around the circular face shared between cylinder top and cone base.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> <a href="#">for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Surface Area of Cylinders (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> <a href="#">for similar readings</a></p> | <p>Learners work through two connected calculations, scaffolded from easier to phenomenon-level: CALCULATION A — Kericho Tea Canister (Cylinder + Cone): Given: cylinder radius <math>r = 5</math> cm, cylinder height <math>h = 12</math> cm, cone slant height <math>l = 7</math> cm, cone base radius = 5 cm (same as cylinder). Learners apply the strategy: (i) Cylinder: full SA = <math>2\pi r(r + h) = 2\pi(5)(5+12) = 170\pi</math> cm<sup>2</sup>. But cylinder top is hidden, so subtract <math>\pi r^2 = 25\pi</math> cm<sup>2</sup>. Exposed cylinder area = <math>145\pi</math> cm<sup>2</sup>. (ii) Cone: full SA = <math>\pi r(r + l) = \pi(5)(5+7) = 60\pi</math> cm<sup>2</sup>. But cone base is hidden, so subtract <math>\pi r^2 = 25\pi</math> cm<sup>2</sup>. Exposed cone area = <math>35\pi</math> cm<sup>2</sup>. (iii) Total composite SA = <math>145\pi + 35\pi = 180\pi \approx 565.5</math> cm<sup>2</sup>. CALCULATION B — Cylinder-Hemisphere Water Tank (Phenomenon): Using dimensions established in prior lessons for a 5,000-litre capacity tank: cylinder radius <math>r = 0.8</math> m, cylinder height <math>h = 2.0</math> m, hemisphere radius <math>r = 0.8</math> m. Learners apply the strategy: (i) Cylinder: lateral surface = <math>2\pi rh = 2\pi(0.8)(2.0) = 3.2\pi</math> m<sup>2</sup>. Cylinder base (bottom, exposed) = <math>\pi r^2 = 0.64\pi</math> m<sup>2</sup>. Cylinder top is hidden</p> | <p>Distribute the worked-example scaffold sheet for Calculation A. Say: 'Work through the Kericho canister calculation in your groups using the three-step strategy from the video. Show every step — identify, subtract, sum. You have 6 minutes.' [Allow 6 minutes.] After 6 minutes, cold-call one group to present each step on the board (3 different groups for steps 1, 2, 3). Correct any errors before moving to Calculation B. Say: 'Now for the real question — the tank from Kisumu. These are the actual dimensions we have been using since Lesson 1. Use the exact same three-step strategy.' [Allow 8 minutes for group work on Calculation B.] Circulate and prompt with: 'Which face is touching the hemisphere flat side? Is it exposed to the outside?' and 'What does the cylinder's bottom face represent in the real tank — is it exposed?' After 8 minutes, say: 'Look at the number you got for the cylinder-hemisphere tank. Now find your frustum surface area from Lesson 4. Write both in your Comparison Table.' [Allow 2 minutes.] Ask: 'Who is surprised by</p> | <p>Scaffolded Parallel Calculation + Comparison Table: Learners first build procedural fluency on a low-stakes Kenya context (Kericho canister), then immediately transfer the identical strategy to the phenomenon context (Kisumu tank). The Comparison Table makes the quantitative contrast between the two tanks explicit and visible, directly advancing the driving question with concrete numerical evidence.</p> | <p>Observable indicator: Learners' Calculation B shows three clearly separated rows (cylinder lateral, cylinder base, hemisphere curved) with the hidden face correctly subtracted and a final answer of approximately 16.09 m<sup>2</sup>, compared to their Lesson 4 frustum surface area. Teacher spot-checks at least 5 groups during independent work. Red flag: learners who include the full cylinder SA formula (<math>2\pi r(r+h)</math>) without subtracting the hidden top — teacher redirects: 'Draw the tank. Put your finger on the flat circle where the hemisphere sits. Can rain or a hand touch that face? No — so it is hidden. Subtract it.'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>(joined to hemisphere flat face). (ii) Hemisphere: curved surface = <math>2\pi r^2 = 2\pi(0.64) = 1.28\pi \text{ m}^2</math>. Hemisphere flat base is hidden. (iii) Total SA = <math>3.2\pi + 0.64\pi + 1.28\pi = 5.12\pi \approx 16.09 \text{ m}^2</math>. Learners then retrieve their frustum surface area from Lesson 4 notes and record both values in a Comparison Table: Tank   Surface Area (<math>\text{m}^2</math>)   Volume (L). They answer: 'Which tank uses more metal? By how much (in <math>\text{m}^2</math>)?' in writing.</p>                                                                                                                                                                                                                                                                                                                                                                                              | <p>what they see? Hands up.' Acknowledge responses. Say: 'This is the mathematical evidence the Grade 10 students at the Kisumu factory were looking for. The numbers are in front of us — what do they tell us?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area of Cylinders Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the class Driving Question Board. The DQB has two anchor columns: 'What do we KNOW?' and 'What do we still WONDER?' Each group receives two sticky notes. On the first (green — KNOW), they write one evidence statement from today's lesson, e.g., 'The cylinder-hemisphere tank has a surface area of <math>\sim 16.09 \text{ m}^2</math> while the frustum tank has <math>\sim 13.2 \text{ m}^2</math> — for the same 5,000-litre volume, the frustum uses less metal.' On the second (yellow — WONDER), they write one new question that today's evidence raises, e.g., 'Is there a shape that would use even less metal for 5,000 litres?' or 'Does the thickness of the metal sheet also affect cost?' Groups post their notes on the DQB. The teacher facilitates a 3-minute gallery read where learners walk past the board silently, then one spokesperson per group shares the most interesting WONDER note.</p> | <p>Say: 'The DQB is where we track what the maths is proving to us. Take your two sticky notes. On the green one, write ONE thing today's calculation proved — use actual numbers. On the yellow one, write ONE new question today's evidence made you think of. You have 3 minutes.' [Allow 3 minutes.] After posting, say: 'Walk past the board — silent gallery read for 90 seconds. Read what other groups wrote.' [Allow 90 seconds.] Then: 'I'm going to cold-call one person from three different groups to share the WONDER question their group posted and explain why today's evidence made them ask it.' Cold-call 3 students. After each, ask the class: 'Does anyone's calculation from today help start to answer this question?' [Allow 10-second wait time after each.] Close by saying: 'Our DQB is building a mathematical case. By the end of this unit, we will be able to answer every question on this board using the mathematics we are learning.'</p> | <p>Driving Question Board Evidence Posting: Requiring learners to express today's findings as an evidence statement with actual numbers (not just 'we learned surface area') forces precise mathematical communication and connects procedural calculation to the conceptual storyline of the phenomenon. The WONDER notes sustain inquiry momentum across the lesson sequence.</p> | <p>Observable indicator: Green sticky notes contain a specific quantitative comparison (surface area values, difference in <math>\text{m}^2</math>) rather than a vague summary. Yellow sticky notes pose a genuine mathematical or engineering question rather than a personal comment. Teacher reads all green notes before the next lesson to identify groups whose surface area value is significantly off — these groups receive targeted feedback before Lesson 6.</p> |
| <p><b>Model Building Phase</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Learners receive their cumulative model diagram — a cross-section drawing of both tanks (cylinder-</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Distribute the cumulative model diagrams (pre-drawn cross-section sheets used since Lesson 1). Say:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Cumulative Annotated Model Revision: The cross-section diagram serves as a running</p>                                                                                                                                                                                                                                                                                           | <p>Observable indicator: Learners' model diagrams show three distinctly coloured and numerically</p>                                                                                                                                                                                                                                                                                                                                                                         |

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| <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Heights of Cylinders Given Surface Area (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>hemisphere and frustum) that they have been annotating since Lesson 1. In this lesson's revision, they: (a) shade or colour-code each distinct surface region of the cylinder-hemisphere tank (cylinder base, cylinder lateral, hemisphere curved) in three different colours; (b) write the calculated area of each region next to it (e.g., cylinder base: <math>0.64\pi \text{ m}^2</math>); (c) draw a double-headed arrow between the two tanks labelled 'Same Volume, Different Surface Area' and annotate the arrow with the numerical difference in surface area; (d) write one sentence at the bottom of their model: 'The cylinder-hemisphere tank requires ___ <math>\text{m}^2</math> more metal than the frustum tank because ___.' Learners share their completed model annotation with their group for peer feedback using the prompt: 'Is every surface region labelled and calculated? Is the hidden face correctly shown as subtracted?'</p> | <p>'Today you are adding the surface area breakdown of the cylinder-hemisphere tank to your model. Use three colours — one for each exposed surface region — and write the area next to each region. Then annotate the double arrow between the tanks. You have 6 minutes.' [Allow 6 minutes.] Circulate and check that learners are writing numerical values, not just labels. After 6 minutes, say: 'Share your model with your group. Use the peer feedback prompt on the board: Is every surface region labelled and calculated? Is the hidden face correctly shown as subtracted? Give your partner 60 seconds of feedback.' [Allow 2 minutes for peer feedback.] Close: 'By the end of Lesson 10, this model will be your complete answer to the driving question. Every number you add today is evidence the Kisumu factory students — and you — can use to explain their observation. Well done.'</p> | <p>scientific model that grows more complete with each lesson. Colour-coding surface regions and writing numerical values alongside them makes the abstract formula results spatially concrete. The comparative annotation ('Same Volume, Different Surface Area') directly addresses the driving question and reinforces the central phenomenon insight.</p> | <p>labelled surface regions for the cylinder-hemisphere tank, with the shared circular face explicitly marked as 'hidden — subtracted.' The double-headed arrow is annotated with the correct numerical surface area difference. Teacher collects 5–6 models at random at the end of the lesson to review before Lesson 6. Red flag: learners who have the correct total but cannot identify which region contributes which area — this indicates procedural rather than conceptual understanding and will need addressing in Lesson 6's review opener.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were learners able to correctly identify the hidden face in the cylinder-hemisphere composite without prompting — or did most groups initially include it in the total? This reveals whether the 'subtract the junction' concept was internalised from the video or needs re-teaching. (2) When learners completed the Comparison Table, what was the emotional or intellectual reaction in the room? Did the numerical difference between the two tanks' surface areas generate surprise, curiosity, or disengagement? This affective response is a key signal of whether the phenomenon is genuinely driving inquiry. (3) How accurate were the Lesson 4 frustum surface area values that learners retrieved from their notes? If many groups had incorrect frustum values, the comparison in this lesson was built on faulty evidence — consider opening Lesson 6 with a brief class consensus on the correct frustum surface area. (4) Did the Kericho canister scaffold (Calculation A) successfully bridge learners from the video to the phenomenon tank (Calculation B), or did most groups stall on Calculation B anyway? If so, consider adding one more intermediate practice example in Lesson 6. (5) Review the green sticky notes from the DQB: how many groups wrote specific numerical evidence versus vague statements? This tells you whether your learners are ready to make quantitative arguments about the phenomenon or still need support translating calculations into claims.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| <b>What did I observe?</b>                   | Learners watched a video demonstration of the three-step composite surface area strategy (identify components, subtract hidden faces, sum exposed surfaces) and examined a labelled diagram of a Kericho tea canister (cylinder + cone) to identify which faces are exposed and which are hidden at the joint.                                                                                                                                                                |
| <b>What did I learn?</b>                     | Learners calculated that the cylinder-hemisphere water tank has a total exposed surface area of approximately $16.09 \text{ m}^2$ ( $5.12\pi \text{ m}^2$ ), which is measurably larger than the frustum tank's surface area computed in Lessons 3–4 — even though both tanks hold exactly 5,000 litres of water. They learned that the composite surface area equals the sum of all component surface areas minus twice the area of each shared (hidden) face at a junction. |
| <b>How does this explain the phenomenon?</b> | Learners explained — using their Comparison Table and annotated cumulative model — that the frustum tank uses less metal sheet than the cylinder-hemisphere tank because its shape produces a smaller total surface area for the same enclosed volume, providing direct mathematical evidence to answer the driving question about why the Kisumu factory's frustum tank is more material-efficient.                                                                          |

## LESSON 6 (80 minutes): Volume of Prisms, Pyramids, Cones, and Spheres

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Purpose</b>              | Learners derive and apply formulae for the volume of prisms, pyramids, cones, and spheres, and use these to compare the storage capacities of differently shaped grain stores — advancing the driving question by establishing that volume and surface area are independent properties of a solid.                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) state the volume formulae for prisms (including cuboids and cylinders), pyramids, cones, and spheres; (b) derive the relationship $V(\text{pyramid}) = \frac{1}{3} \times V(\text{prism})$ with the same base and height; (c) calculate the volume of composite solids involving combinations of these shapes; (d) compare volumes of differently shaped solids to explain observed differences in storage capacity.                                                                                                                                                                                                           |
| <b>Skills</b>               | By the end of the lesson, the learner should be able to: (a) select and correctly apply the appropriate volume formula for a given solid; (b) perform multi-step volume calculations with correct units ( $\text{cm}^3$ , $\text{m}^3$ , litres); (c) convert between volume units ( $1 \text{ m}^3 = 1\,000 \text{ litres}$ ); (d) construct and revise a cumulative class model linking solid shape to volume and surface area.                                                                                                                                                                                                                                                           |
| <b>Attitudes</b>            | Learners appreciate that mathematical formulae are practical tools used by Kenyan engineers, grain-store designers, and water-tank manufacturers to make cost-effective decisions, and take responsibility for the accuracy of their calculations as a form of professional integrity.                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Key Inquiry Question</b> | How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Purpose in Storyline</b> | In Lesson 6, learners shift the investigative lens from surface area (Lessons 3–5) to volume. Using the Kitale maize storage supporting phenomenon, they compute and compare the volumes of a rectangular grain store (prism) and a conical grain store, explaining why a farmer notices that two structures of the same height and base width store different amounts of maize. This finding feeds directly back to the Kisumu tank anchoring phenomenon: both the cylindrical-hemispherical tank and the frustum tank hold 5 000 litres — learners now have the mathematical tools to verify that claim and to understand how manufacturers engineer equal volumes into different shapes. |
| <b>Safety Notes</b>         | No hazardous materials are used. Where physical containers of water or sand are used for demonstration, ensure surfaces are dry to prevent slipping. Handle any glass measuring cylinders with care.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### B. LESSON OVERVIEW

Learners begin by predicting which of two grain stores — a rectangular prism store and a conical store of identical base and height dimensions — holds more maize, connecting to the Kitale supporting phenomenon. Through guided derivation and a hands-on sand/water displacement activity, they establish volume formulae for prisms, pyramids, cones, and spheres, discovering the  $\frac{1}{3}$  relationship between cone/pyramid and prism/cylinder volumes. They then calculate, compare, and verify volumes of the two tank designs from the Kisumu anchoring phenomenon, confirming the manufacturer's claim of equal 5 000-litre capacity. The lesson closes with model revision on the class Driving Question Board, where learners articulate that shape affects surface area and volume independently — a key disciplinary insight.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>        | <p>Learners examine a labelled diagram (projected or printed) showing two grain stores used by a farmer in Kitale, Trans-Nzoia County: Store A is a rectangular prism (length 3 m, width 3 m, height 4 m) and Store B is a cone (base radius 1.5 m, height 4 m). Both stores have the same base width (3 m diameter / 3 m side) and the same height (4 m). Working individually for 3 minutes, learners write a prediction in their exercise books: 'Which store holds more maize — Store A, Store B, or do they hold the same amount? Give a reason for your prediction.' They then share predictions with a shoulder partner (1 minute) before a whole-class show-of-hands poll.</p>                             | <p>Display the two grain-store diagrams on the board with all dimensions clearly labelled. Say: 'A farmer in Kitale has built two grain stores — both are the same height and the base of each store fits inside the same 3-metre square footprint. He is puzzled: one store seems to hold much more maize than the other. Before we calculate anything, I want your best scientific prediction.' Circulate silently for 3 minutes — do NOT answer questions about formulae. After the shoulder-partner share, conduct a cold-call poll: call on 4 learners (at least one from each prediction camp — Store A, Store B, same). Ask each: 'What reasoning led you to that prediction?' Apply 10-second wait time before accepting responses. Record tally of predictions on the board (Store A / Store B / Same) and say: 'Let us hold these predictions — our calculations will tell us who was right and why.'</p> | <p>Prediction Anchoring — by committing to a written prediction before instruction, learners activate prior intuitions about shape and space. The shoulder-partner share surfaces diverse reasoning (e.g., 'a cone tapers so it must hold less' vs. 'they have the same height so it should be the same'), creating productive cognitive conflict that motivates the derivation activity.</p>                                                                                                                                                                                                  | <p>Observable indicator: Every learner has a written prediction with a stated reason in their exercise book before the poll. Teacher scans books during circulation to check reasoning quality. Key diagnostic: learners who say 'same amount because same height and base' reveal a misconception about how shape affects volume — flag these learners for targeted questioning during Phase 3.</p>                                                                                                                                                                                |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Activity 2A — Hands-on Derivation (Cone vs. Cylinder): In groups of four, learners receive a cylindrical container and a conical container of identical base radius and height (e.g., a tin cylinder and a paper cone made from the template provided, both with <math>r = 5</math> cm, <math>h = 10</math> cm). Groups fill the cone with dry sand, then pour it into the cylinder, repeating until the cylinder is full. Learners record: 'How many full cones of sand fill the cylinder?' They record the result (3 cones = 1 cylinder) in a results table. Activity 2B — Formula Derivation Reading: Learners read a structured half-page text (provided as a printed card or displayed on screen) that</p> | <p>For Activity 2A: Distribute materials and say: 'Your task is not to guess — measure. Fill the cone, pour it into the cylinder, count how many times. Record the number accurately.' Monitor all groups; after 5 minutes, ask: 'Groups, what number did you get?' Cold-call 3 groups. Write results on the board. Say: 'What mathematical relationship does this suggest between the volume of a cone and the volume of a cylinder with the same base and height?' Wait 12 seconds. Cold-call 2 learners. Confirm: 'Yes — the cone holds exactly one-third of the cylinder. This is not a coincidence — it is a geometric truth we can prove with calculus later in your schooling,</p>                                                                                                                                                                                                                           | <p>Concrete-Representational-Abstract (CRA) Progression — learners first handle physical objects (cone and cylinder with sand) to observe the <math>\frac{1}{3}</math> relationship concretely, then read its symbolic representation in the formula card, and finally apply it abstractly in calculation. This three-step sequence builds durable conceptual understanding rather than rote formula memorisation, directly aligned with NGSS Science and Engineering Practice 3 (Planning and Carrying Out Investigations) and Practice 5 (Using Mathematics and Computational Thinking).</p> | <p>Observable indicators: (1) Every group correctly records '3 cones = 1 cylinder' (or equivalent) in their results table — groups recording other values need immediate re-measurement support. (2) In pair calculations, Store A volume = <math>36 \text{ m}^3 = 36\,000</math> litres; Store B volume = <math>\frac{1}{3} \times \pi \times 1.5^2 \times 4 \approx 9.42 \text{ m}^3 \approx 9\,420</math> litres. Teacher checks at least 6 pairs' working during circulation. Flag any pair that omits the <math>\frac{1}{3}</math> factor or forgets to square the radius.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>presents the volume formulae for a prism (<math>V = \text{Base Area} \times \text{height}</math>), pyramid (<math>V = \frac{1}{3} \times \text{Base Area} \times \text{height}</math>), cylinder (<math>V = \pi r^2 h</math>), cone (<math>V = \frac{1}{3} \pi r^2 h</math>), and sphere (<math>V = \frac{4}{3} \pi r^3</math>), with a worked example for each using Kenyan contexts (a cylindrical water tank in Kisumu, a conical grain store in Kitale, a spherical gas container in Nairobi). Activity 2C — Worked Calculation Pair Task: Pairs work through two calculations: (1) Volume of Store A (rectangular prism, <math>3 \times 3 \times 4</math> m); (2) Volume of Store B (cone, <math>r = 1.5</math> m, <math>h = 4</math> m). They record answers in <math>\text{m}^3</math> and convert to litres.</p>                                                                                                                                                                                                                                                                                      | <p>but today we will use it as our working rule.' For Activity 2B: Say: 'Read the formula card carefully. As you read, underline the <math>\frac{1}{3}</math> factor wherever it appears and ask yourself: which solids share this factor and why?' For Activity 2C: Circulate during pair work. When a pair finishes, ask: 'How did you convert your answer from <math>\text{m}^3</math> to litres?' (Target answer: multiply by 1 000.) If a pair struggles with the cone formula, prompt: 'What is the base shape of the cone? What is its area? Now multiply by height and then by...?' Wait 10 seconds.</p>                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to their Phase 1 predictions and self-assess in writing: 'Was your prediction correct? What does the calculation tell you about how shape affects volume?' (2 minutes individual writing.) Groups of four then tackle the Explain Challenge — a three-part problem printed on a task card: Part 1: A farmer in Kitale claims his rectangular grain store (<math>3 \text{ m} \times 3 \text{ m} \times 4 \text{ m}</math>) holds nearly 4 times more maize than his neighbour's conical store (<math>r = 1.5</math> m, <math>h = 4</math> m). Use your calculations from Phase 2 to verify or refute this claim and express the ratio of their volumes. Part 2: A spherical metal gas container in a Nairobi industrial park has a radius of 2 m. Calculate its volume in <math>\text{m}^3</math>. Part 3 (Extension — connects back to anchoring phenomenon): The Kisumu tank manufacturer claims both tanks hold exactly 5 000 litres. The cylindrical-hemispherical tank has a cylinder of radius 0.7 m and height 2.8 m, topped with a hemisphere of radius 0.7 m. Calculate the total</p> | <p>Open Phase 3 by returning to the prediction tally on the board: 'Let us see what our evidence says.' Cold-call one learner from each prediction group to read their self-assessment. Affirm accurate reasoning and gently correct misconceptions: 'Store B is a cone — it tapers to a point, so even though it has the same base and height, it only fills one-third of the space a cylinder would fill. That is why shape matters so much.' Distribute the Explain Challenge task cards. For Part 3, scaffold with: 'What two shapes make up the composite tank? Write the formula for each, then add them.' During the gallery walk, pose probing questions to each group: 'What does your answer in litres tell you about the manufacturer's claim?' 'If your answer is not exactly 5 000 litres, what might explain the difference?' Apply 12-second wait time consistently. After the gallery walk, deliver a 5-minute direct teaching summary on the board, writing all four formulae clearly with their Kenyan</p> | <p>Claim–Evidence–Reasoning (CER) Writing — learners write a conclusion sentence for each task part following the structure: Claim (what the answer shows), Evidence (the calculated values), Reasoning (the formula and mathematical principle applied). This strategy, aligned with NGSS Science and Engineering Practice 6 (Constructing Explanations), trains learners to communicate mathematical meaning rather than just produce numerical answers, building the analytical writing skills required for KCSE examination responses.</p> | <p>Observable indicators: (1) Part 1 — learners correctly compute the volume ratio: <math>36 \div 9.42 \approx 3.82</math>, confirming the farmer's claim is approximately correct. Accept answers between 3.7 and 3.9. (2) Part 2 — <math>V(\text{sphere}) = \frac{4}{3} \times \pi \times 2^3 \approx 33.51 \text{ m}^3</math>. (3) Part 3 — <math>V(\text{cylinder}) = \pi \times 0.7^2 \times 2.8 \approx 4.31 \text{ m}^3</math>; <math>V(\text{hemisphere}) = \frac{2}{3} \times \pi \times 0.7^3 \approx 0.718 \text{ m}^3</math>; Total <math>\approx 5.03 \text{ m}^3 \approx 5\,030</math> litres — close to 5 000 litres (manufacturer's claim is approximately valid). Teacher uses these expected answers as a diagnostic rubric during the gallery walk.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>volume of this composite tank in litres and assess whether the manufacturer's claim is accurate. Groups record full working and a written conclusion sentence. A spokesperson from each group shares one part of the solution in a class gallery walk.</p>                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>context examples, and explicitly stating: 'Volume tells us capacity — how much a solid can hold. Surface area tells us material — how much it costs to build. These two properties are independent of each other — that is the central mathematical truth of this entire unit.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners return to the class Driving Question Board displayed at the front of the room. Each group receives two sticky notes of different colours: a GREEN note ('Evidence we now have') and an ORANGE note ('New questions this raises'). Groups write one entry for each colour based on today's lesson findings and post them on the DQB under the relevant driving question column. Individually, learners also update their personal DQB tracker in their exercise books, completing the sentence stems: 'Today I found evidence that... [connect to volume formulae and Kitale phenomenon]' and 'This changes/confirms my thinking about the Kisumu tanks because...'</p> | <p>Say: 'We have now investigated surface area across five lessons and today we have explored volume. The Driving Question asks how shape determines both material needed AND how much a solid can hold. What new evidence can we post today?' Give groups 3 minutes to write their sticky notes. As groups post, read 2–3 aloud and ask the class: 'Does everyone agree this counts as evidence? Does it directly address our driving question?' Cold-call 2 learners to read their personal tracker entries. Highlight for the class: 'Notice that today we confirmed the manufacturer's claim — the cylindrical-hemispherical tank does hold approximately 5 000 litres. In the next lesson, we will investigate the frustum tank to see if that claim also holds, and then we can compare their surface areas to explain why one uses less metal.'</p> | <p>Driving Question Board Protocol — the structured posting of evidence and new questions makes the cumulative nature of the inquiry visible to all learners. The colour-coded system distinguishes what is known (green) from what remains to be investigated (orange), modelling authentic scientific practice (NGSS Practice 1: Asking Questions) and helping learners track the arc of understanding across all 10 lessons.</p> | <p>Observable indicators: Green sticky notes must contain a specific mathematical finding (e.g., 'The cone holds only <math>\frac{1}{4}</math> of what the rectangular prism holds for the same base and height'). Orange sticky notes must contain a genuine investigative question connected to the phenomenon (e.g., 'Does the frustum tank also hold 5 000 litres? We need to check!'). Teacher reads all posted notes after class to identify misconceptions for addressing in Lesson 7.</p> |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b></p>                                                                                                                                                                                                                                                    | <p>Learners individually revise and extend their cumulative explanatory model — a labelled diagram or annotated sketch they have been building since Lesson 1. Today's revision task: (1) Add a 'Volume' column or panel to their model alongside the existing 'Surface Area' panel. (2) Annotate at least three solids (cylinder, cone, rectangular prism) with both their surface area formula and</p>                                                                                                                                                                                                                                                                           | <p>Say: 'Your model is your growing scientific explanation of how shape, surface area, and volume are related. Today you discovered something powerful: volume and surface area are independent. Your model must show this.' Project a partially completed exemplar model on the board (teacher-prepared) showing the structure but with blank formula spaces, so learners see the format</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Cumulative Model Revision — iterative model building across lessons mirrors authentic scientific and engineering practice (NGSS Practice 2: Developing and Using Models). By physically adding the volume panel alongside the surface area panel, learners create a visual representation of the independence of these two properties, making the abstract mathematical relationship</p>                                         | <p>Observable indicators: (1) The model contains correctly labelled volume formulae for at least three solids. (2) The 'Independent properties' arrow is present and the learner can verbally explain what it means if cold-called. (3) The closing sentence accurately connects the mathematical finding to the Kisumu anchoring phenomenon. Teacher collects 5 randomly selected models at the</p>                                                                                              |

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| <a href="#">Apply volume of solids</a><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES</a> for similar readings | their volume formula side by side. (3) Draw an arrow labelled 'Independent properties' between the Surface Area panel and the Volume panel to represent today's central insight. (4) Write one sentence at the bottom of the model: 'A manufacturer in Kisumu can engineer equal volumes into different shapes, but the surface area — and therefore the material cost — will differ depending on the shape chosen.' The class shares models in a 3-minute pair comparison before the lesson closes. | without copying content. Circulate during the 7-minute model revision, prompting: 'Does your model show that the cone and cylinder have the same base and height but different volumes?' and 'Have you connected your model back to the Kisumu phenomenon yet?' After the pair share, cold-call 2 learners to read their closing sentence aloud. Close the lesson by saying: 'In Lesson 7, we will calculate the volume of a frustum — the shape of the second Kisumu tank — and then we will have everything we need to fully explain the factory observation that started our entire investigation.' | cognitively tangible and retrievable. | end of class to assess depth of understanding before Lesson 7. |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. PREDICTION ACCURACY: What proportion of learners correctly predicted that Store A (rectangular prism) holds more maize than Store B (cone)? What reasoning patterns did incorrect predictors use — and how will you address these in Lesson 7's opener?
2. HANDS-ON DERIVATION: Did all groups successfully observe the 3:1 cylinder-to-cone relationship in Activity 2A? If groups got results like 2.5 or 3.5, what measurement errors occurred — and would a digital simulation be a more reliable alternative for your classroom context?
3. COMPOSITE VOLUME (PART 3): How many groups successfully calculated the composite cylinder-hemisphere volume and connected it to the manufacturer's 5 000-litre claim? Groups that could not attempt Part 3 may need additional scaffolding on composite solids before Lesson 8 (which addresses the frustum).
4. DQB QUALITY: Review the sticky notes posted on the Driving Question Board. Do the green notes contain specific mathematical evidence, or are they vague ('we learned about volume')? Plan targeted prompting for Lesson 7 to raise the precision of DQB contributions.
5. MODEL PROGRESS: From the 5 models collected, did learners correctly represent the independence of surface area and volume? If most models show the two properties as connected (e.g., 'bigger surface area = bigger volume'), this is a critical misconception to address explicitly at the start of Lesson 7 using a counterexample — such as comparing a flat sheet (large surface area, small volume) with a sphere (compact shape, efficient volume).
6. PACING: The lesson has a high activity density (sand activity + reading + three-part calculation + DQB + model). If pacing was tight, consider assigning Part 3 of the Explain Challenge as a homework task reviewed at the start of Lesson 7, freeing time for deeper model-building in class.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| What did I observe? | Learners observed that filling a cone with sand and pouring it into a cylinder of equal base and height requires exactly 3 cones to fill the cylinder — providing concrete, physical evidence that the volume of a cone is one-third the volume of its corresponding cylinder. They also observed labelled diagrams of two grain stores (a rectangular prism and a cone) in Kitale, Trans-Nzoia County, with identical base widths and heights. |
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| <b>What did I learn?</b>                     | Learners learned the volume formulae for prisms ( $V = \text{Base Area} \times h$ ), pyramids and cones ( $V = \frac{1}{3} \times \text{Base Area} \times h$ ), cylinders ( $V = \pi r^2 h$ ), and spheres ( $V = \frac{4}{3} \pi r^3$ ). They discovered that the $\frac{1}{3}$ factor arises because a cone or pyramid occupies exactly one-third of the space of a prism or cylinder with the same base and height. They learned to convert between $\text{m}^3$ and litres ( $1 \text{ m}^3 = 1\,000 \text{ L}$ ) and calculated that the cylindrical-hemispherical Kisumu tank holds approximately 5 030 litres — consistent with the manufacturer's 5 000-litre claim. |
| <b>How does this explain the phenomenon?</b> | Learners explained that the Kitale farmer's rectangular grain store holds approximately 3.8 times more maize than the conical store with the same base and height, because the cone's tapering shape means it occupies only one-third of an equivalent cylinder's volume. They further explained that the Kisumu tank manufacturer can engineer equal volumes (5 000 litres) into very different shapes — cylindrical-hemispherical vs. frustum — confirming that volume and surface area are mathematically independent properties, and that shape choice affects material cost (surface area) without necessarily changing storage capacity (volume).                      |

## LESSON 7 (80 minutes): Volume of Frustums of a Cone

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | To enable learners to derive and apply the formula for the volume of a frustum of a cone in real-world contexts, specifically to verify the storage capacity of the frustum-shaped water tank from the anchoring phenomenon.                                                                                                                                                                                                                                                                                 |
| <b>Knowledge</b>            | Learners will know that the volume of a frustum is derived by subtracting the volume of the smaller removed cone from the volume of the original larger cone, expressed as $V = (1/3)\pi h(R^2 + Rr + r^2)$ , where R and r are the radii of the larger and smaller circular faces respectively and h is the perpendicular height of the frustum.                                                                                                                                                            |
| <b>Skills</b>               | Learners will be able to: (a) derive the frustum volume formula from the difference of two cones; (b) identify the dimensions R, r, and h from a given frustum; (c) calculate the volume of a frustum using the derived formula; (d) convert between cubic centimetres/metres and litres to verify real-world capacity claims.                                                                                                                                                                               |
| <b>Attitudes</b>            | Learners will appreciate how mathematical derivation provides tools that engineers and manufacturers in Kenya use to make evidence-based design decisions that balance cost and functionality.                                                                                                                                                                                                                                                                                                               |
| <b>Key Inquiry Question</b> | How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | In Lesson 6 learners calculated the surface area of the frustum tank and confirmed it uses less metal than the composite cylinder-hemisphere tank. A key question remained unanswered on the DQB: Does the frustum tank truly hold 5,000 litres? Lesson 7 equips learners with the volume formula for a frustum so they can compute the internal volume of the Kisumu factory's frustum tank, verify the manufacturer's 5,000-litre claim, and move one step closer to fully resolving the driving question. |
| <b>Safety Notes</b>         | No hazardous materials are used. If learners handle physical cone/frustum cut models, ensure edges are not sharp. Devices used to stream or view the YouTube video should be handled responsibly and returned to the charging station after use.                                                                                                                                                                                                                                                             |

### B. LESSON OVERVIEW

Learners open by revisiting their DQB and recalling that the frustum tank's surface area is smaller than the composite tank's — but they have not yet verified the volume claim. Through a short prediction task, learners estimate how the volume of a frustum relates to the two cones involved. They then watch a curated YouTube video demonstrating the derivation of the frustum volume formula from the subtraction of cone volumes. Using structured notes, learners work through the algebraic derivation in pairs. They apply the formula to the Kisumu factory's frustum tank dimensions to confirm whether it holds 5,000 litres ( $\approx 5 \text{ m}^3$ ), recording findings as evidence. The lesson closes with a DQB update and a cumulative model revision that now includes both surface area and volume comparisons for the frustum tank.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                      | Learner Experience                                                                                                                     | Teacher Moves                                                                                                                     | Sensemaking Strategy                                                                                                     | Formative Assessment Strategy                                                                                                        |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Determining similar triangles</a> | Learners individually study a labelled cross-section diagram of a frustum showing the original large cone (height H, radius R) and the | Display the cross-section diagram on the board or screen. Say verbatim: 'Before we derive any formula today, I want your brain to | Prediction Journaling — by committing a written prediction before instruction, learners activate prior knowledge of cone | Observable indicator: Each learner has a written prediction with a justification referencing cone volume within the first 3 minutes. |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                               | <p>small removed cone (height <math>h_s</math>, radius <math>r</math>). They respond in their exercise books to the prompt: 'The Kisumu factory's frustum tank is made by cutting the top off a large cone. Without calculating, predict: is the volume of the frustum closer to the volume of the big cone, exactly half of it, or something else? Write a sentence justifying your prediction using what you already know about cone volumes.' After 3 minutes of silent individual thinking, learners share predictions with a shoulder partner for 90 seconds.</p>                                                                                                                                                                                                                                             | <p>commit to a prediction — this is how real engineers in Kisumu think before they calculate.' Circulate and read 3–4 exercise books silently. After partner sharing, cold-call 3 learners using a random name selector or row-by-row system. Ask each: 'What did you predict and why?' Allow 10 seconds of WAIT TIME after each response before commenting. Do NOT confirm or deny predictions — say: 'Hold that thought. The video we are about to watch will either confirm or challenge you.'</p>                                                                                                                                                                             | <p>volume (<math>V = \frac{1}{3}\pi r^2 h</math>) and create a cognitive anchor. When the derivation later confirms or refines their prediction, the contrast drives deeper encoding of the frustum formula.</p>                                                                                                                              | <p>Teacher scans for misconceptions — e.g., learners who think the frustum volume equals the full cone volume (forgetting subtraction) or who cannot recall the cone volume formula. Note these learners for targeted support during the Observe phase.</p>                                                                                                                                                                    |
| <p><b>Observe Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Solving similar triangles</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners watch a teacher-selected YouTube video (approximately 8–10 minutes) demonstrating the derivation of the frustum volume formula. Before the video begins, each learner receives a structured note-taking frame with three columns: 'What I see/hear', 'Mathematical step', 'Question I have'. Learners pause the video at two teacher-designated timestamps — (1) after the setup of the two cones and (2) after the algebraic simplification — to discuss with their partner what they wrote. After the video, learners individually complete a 'Video Summary Strip': they write the frustum volume formula in their own words and label a blank frustum diagram with <math>H</math>, <math>h_s</math>, <math>R</math>, <math>r</math>, and <math>h</math> (perpendicular height of the frustum).</p> | <p>Before playing the video say verbatim: 'You are watching this video the way the Grade 10 students in Kisumu would watch a training video at the factory — with a purpose. Your job is to track every mathematical move the presenter makes.' Pause the video at the two designated timestamps. At Pause 1 ask: 'What are the two cones the presenter identified? Write their dimensions.' At Pause 2 ask: 'What operation connects the two cone volumes? Why subtraction?' Cold-call 2 learners at each pause — allow 10 seconds of WAIT TIME. After the video, circulate and check that every learner's diagram is correctly labelled before moving to the Explain phase.</p> | <p>Structured Note-Taking with Designated Pauses — the three-column frame prevents passive watching by requiring simultaneous processing. The two pauses create retrieval checkpoints that consolidate partial understanding before the derivation continues, mimicking the 'chunking' technique used in engineering training programmes.</p> | <p>Observable indicator: Learner's Video Summary Strip shows the correct formula <math>V_{\text{frustum}} = (1/3)\pi h(R^2 + Rr + r^2)</math> and a correctly labelled frustum diagram. Teacher checks 6–8 strips during the partner discussion pauses. Red flag: learners who write <math>V = (1/3)\pi R^2 H - (1/3)\pi r^2 h_s</math> but cannot simplify further — these learners need the algebraic bridge in Phase 3.</p> |
| <p><b>Explain Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Inscribed Similar Triangles - Basic Principles</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar videos</a></p>                                                                                                                                                                                             | <p>Learners work in pairs through a three-part structured worksheet. PART A — Derivation Reconstruction: Given the expressions <math>V_{\text{big}} = (1/3)\pi R^2 H</math> and <math>V_{\text{small}} = (1/3)\pi r^2 h_s</math>, and the similar-triangles relationship</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>For Part A, write each algebraic step on the board one line at a time. Say: 'Before I write the next line, complete it in your book. You have 45 seconds.' Use WAIT TIME of 45 seconds consistently. Cold-call one pair per step to read their</p>                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Derivation Reconstruction with Algebraic Chaining — rather than presenting the formula as a given, learners rebuild it step-by-step using the subtraction of cones and similar-triangles reasoning. This strategy connects the frustum</p>                                                                                                 | <p>Observable indicator for Part B: Learner correctly calculates <math>V \approx (1/3) \times \pi \times 2.1 \times (0.81 + 0.45 + 0.25) = (1/3) \times \pi \times 2.1 \times 1.51 \approx 3.328 \text{ m}^3 \approx 3,328 \text{ L}</math> — which is less than 5,000 L, prompting a rich discussion about whether the</p>                                                                                                    |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p><math>H/R = h_s/r</math> (therefore <math>h_s = Hr/R</math>), learners substitute and simplify to derive <math>V_{\text{frustum}} = (1/3)\pi h(R^2 + Rr + r^2)</math>, where <math>h = H - h_s</math>. Teacher guides the class through the algebraic steps on the board, pausing for pairs to complete each line before revealing the next.</p> <p>PART B — Worked Example (Kisumu Factory Tank): The frustum tank has dimensions: <math>R = 0.9</math> m, <math>r = 0.5</math> m, <math>h = 2.1</math> m. Learners calculate its volume and convert to litres (<math>1 \text{ m}^3 = 1,000</math> L) to verify whether it holds 5,000 L.</p> <p>PART C — Extension: A posho mill owner in Kericho stores maize flour in a frustum-shaped silo with <math>R = 1.2</math> m, <math>r = 0.7</math> m, <math>h = 3.0</math> m. Calculate the volume in <math>\text{m}^3</math> and the mass of flour it can hold if the density of maize flour is approximately 600 <math>\text{kg/m}^3</math>.</p> | <p>line aloud — rotate through the room systematically. If a pair gives an incorrect step, say: 'Interesting — can you tell me where you started from?' rather than correcting immediately. For Part B, after pairs calculate, ask: 'Hands up if your answer is between 4,900 L and 5,100 L.' Count hands aloud and record on the board. Then cold-call one pair with the correct answer to narrate their full working. For Part C, circulate to the 3–4 pairs identified during the Observe phase and provide a scaffolded hint card showing the density formula.</p>                                                                                                               | <p>formula back to prior knowledge (cone volume, similarity) and prevents rote memorisation. The Kisumu factory context keeps every algebraic step tethered to the anchoring phenomenon.</p>                                                                                                                                                                                                 | <p>manufacturer's claim is accurate or whether additional dimensions are needed. Teacher notes which pairs reach this conclusion independently as evidence of high-order reasoning.</p>                                                                                                                                                                                    |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">Solving similar triangles: same side plays different roles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each pair receives one sticky note or card. They write one statement beginning with 'We now know...' about the frustum tank's volume and one question beginning with 'We still wonder...' about the driving question. Pairs post their cards on the class DQB under two columns: 'VOLUME EVIDENCE' and 'REMAINING QUESTIONS'. The class reads the board silently for 60 seconds, then the teacher facilitates a 5-minute whole-class discussion connecting today's volume finding to the surface area evidence from Lesson 6. Learners update their individual DQB tracking sheets, marking Lesson 7 as complete.</p>                                                                                                                                                                                                                                                                                                                                                                             | <p>Stand beside the DQB and read two or three 'We still wonder' cards aloud. Say verbatim: 'These questions are gold — they tell us exactly what we still need to investigate. Look at the board: we now know both the surface area AND the volume of the frustum tank. What are we still missing to fully answer our driving question?' Allow 10 seconds of WAIT TIME. Cold-call 3 learners. Guide discussion toward: (1) the composite cylinder-hemisphere tank's volume has not yet been calculated, and (2) a direct numerical comparison of both surface areas and both volumes is still needed. Say: 'Add those to your tracking sheet as the target for Lessons 8 and 9.'</p> | <p>DQB Evidence Mapping — posting findings under 'VOLUME EVIDENCE' makes the accumulation of evidence across lessons visible and spatial. When learners see that Lessons 6 and 7 together now cover surface area AND volume of the frustum, they can identify the remaining gap (the composite tank's volume) as the next investigative target, reinforcing scientific reasoning habits.</p> | <p>Observable indicator: Each pair's 'We now know...' statement correctly references a volume value (in litres or <math>\text{m}^3</math>) for the frustum tank and connects it to the driving question. Teacher checks that at least 80% of pairs have identified 'the composite tank's volume' as a remaining question, confirming readiness to proceed to Lesson 8.</p> |
| <p><b>Model Building</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Learners open their cumulative 'Tank Comparison Model' — a</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Circulate and read the 'Explanation' column entries as</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Cumulative Model Revision — the Tank Comparison Model functions</p>                                                                                                                                                                                                                                                                                                                       | <p>Observable indicator: The learner's model now contains: (a) correctly</p>                                                                                                                                                                                                                                                                                               |

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| <p><b>Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving similar triangles - dubbed(KY)</a><br/> Source: Khan Academy (Kiswahili)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cones Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>table and annotated diagram they have been building since Lesson 1. Today they add a new row to the Volume section of the table for the frustum tank, entering the value calculated in Phase 3. They also annotate their frustum diagram with the formula <math>V = (1/3)\pi h(R^2 + Rr + r^2)</math> and the derived value in litres. In the 'Explanation' column of their model, they write one sentence connecting volume to the anchoring phenomenon: 'The frustum tank's volume of approximately _____ litres [confirms / does not confirm] the manufacturer's 5,000-litre claim because...'. Learners who finish early begin a side-by-side sketch comparing the frustum and composite tanks, labelling all known dimensions and calculated values from Lessons 1–7.</p> | <p>learners write. Look for three quality indicators: (1) the correct volume value is cited, (2) the word 'because' is followed by mathematical reasoning, and (3) the phenomenon (Kisumu factory, 5,000-litre claim) is explicitly named. If a learner's explanation is vague, prompt: 'You said the tank holds less than 5,000 litres — what does that mean for the manufacturer's claim? What would a factory engineer in Kisumu need to check?' Collect 4–5 model sheets to review after class and return with written feedback before Lesson 8.</p> | <p>as a running scientific model that grows in explanatory power with each lesson. Adding the frustum volume today alongside the surface area from Lesson 6 allows learners to see, in one visual artefact, how shape affects both properties simultaneously. This mirrors how engineers maintain and update technical drawings as new data is gathered.</p> | <p>labelled frustum dimensions (R, r, h); (b) the formula written in full; (c) a calculated volume value with unit conversion shown; (d) a written explanation linking the value to the manufacturer's claim. Teacher uses these four criteria as a checklist when reviewing collected sheets. Learners missing criterion (d) are flagged for a brief individual check-in at the start of Lesson 8.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your notes before Lesson 8:

1. DERIVATION DEPTH — Did the majority of pairs successfully reconstruct the algebraic derivation in Part A, or did many learners copy the steps without understanding the similar-triangles link? If the latter, plan a 5-minute retrieval starter for Lesson 8 that re-asks: 'Why is  $h_s/r = H/R$ '?
2. VOLUME RESULT — When learners calculated the frustum tank's volume and found it was significantly less than 5,000 litres using the given dimensions, how did they respond? Did they question the dimensions provided, or did they accept the result uncritically? Use this as a diagnostic of mathematical confidence and critical thinking.
3. DQB QUALITY — Were the 'We still wonder' cards specific enough to drive Lesson 8? If most cards were vague (e.g., 'we wonder about the other tank'), prompt more precise questioning at the start of Lesson 8: 'What specific calculation do we need to do next, and which formula will we use'?
4. PACING — The algebraic derivation in Phase 3 is the most cognitively demanding section. If the class ran short on time, Phase 5 model building can begin as a homework task, with learners completing the 'Explanation' column at home and sharing at the start of Lesson 8.
5. DIFFERENTIATION — Note which pairs required the scaffolded hint card for Part C. Plan to seat these learners near the board during Lesson 8's composite volume derivation so they receive more proximal support.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                     |                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| What did I observe? | Learners watched a YouTube video showing how the volume of a frustum is derived by subtracting the smaller removed cone's |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|

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|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | volume from the larger original cone's volume, pausing at two designated points to record and discuss the mathematical steps.                                                                                                                                                                                                                                                            |
| <b>What did I learn?</b>                     | The volume of a frustum of a cone is given by $V = (1/3)\pi h(R^2 + Rr + r^2)$ , derived from the difference of two cone volumes using similar-triangle proportionality. Learners applied this formula to calculate the volume of the Kisumu factory's frustum-shaped water tank and converted the result to litres to evaluate the manufacturer's 5,000-litre storage claim.            |
| <b>How does this explain the phenomenon?</b> | Learners explained, in their cumulative Tank Comparison Models, how the calculated volume of the frustum tank either confirms or challenges the manufacturer's advertised capacity of 5,000 litres, and identified that the composite cylinder-hemisphere tank's volume must still be calculated before a full comparison — and a complete answer to the driving question — can be made. |

## LESSON 8 (80 minutes): Volume of Frustums of a Pyramid

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Purpose</b>              | To enable learners to derive and apply the formula for the volume of a frustum of a pyramid in real-world contexts related to construction and agribusiness in Kenya.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to derive the formula for the volume of a frustum of a pyramid using the subtraction of pyramid volumes, and apply it to calculate the volume of frustum-shaped structures such as grain stores and architectural bases.                                                                                                                                                                                                                                                                                        |
| <b>Skills</b>               | By the end of the lesson, the learner should be able to: (a) derive the formula $V = (h/3)(A_1 + A_2 + \sqrt{A_1A_2})$ for the volume of a frustum of a pyramid; (b) calculate the volume of frustum-shaped solids given relevant dimensions; (c) solve real-life problems involving frustum grain stores and construction structures in Kenya.                                                                                                                                                                                                                      |
| <b>Attitudes</b>            | The learner appreciates how mathematical reasoning about frustum volumes supports efficient and cost-effective design decisions made by engineers, architects, and agribusiness professionals in Kenya.                                                                                                                                                                                                                                                                                                                                                              |
| <b>Key Inquiry Question</b> | How do we apply surface area and volume of solids in day-to-day activities? How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Purpose in Storyline</b> | In Lesson 7, learners investigated the volume of a frustum of a cone. In Lesson 8, learners extend this understanding to frustums of pyramids — the shape used in many traditional grain stores (mapunjas) and modern architectural plinths across Kenya. By deriving and applying the pyramid-frustum volume formula, learners gather new evidence to compare the frustum-tank design at the Kisumu factory with other frustum-shaped containers, further advancing their answer to the driving question about why shape affects material use and storage capacity. |
| <b>Safety Notes</b>         | No physical hazards anticipated. Ensure all learners have access to a calculator. If the YouTube video is played, confirm the audio volume is appropriate for the classroom setting and that the video has been pre-screened for content suitability.                                                                                                                                                                                                                                                                                                                |

### B. LESSON OVERVIEW

Learners begin by recalling the cone-frustum volume formula from Lesson 7 and predicting whether a similar approach will work for pyramid frustums. They then watch a curated YouTube video demonstrating the derivation of the pyramid-frustum volume formula. Through guided sensemaking, learners derive the formula  $V = (h/3)(A_1 + A_2 + \sqrt{A_1A_2})$  by subtracting the volume of the smaller removed pyramid from the larger whole pyramid. They apply the formula to two Kenya-context problems: (1) a frustum-shaped grain store (mapunjas) used by a maize farmer in the Rift Valley, and (2) a frustum-shaped concrete base for a water tower in Siaya County — directly echoing the anchoring phenomenon. Learners update their Driving Question Boards and revise their cumulative models to incorporate pyramid-frustum volume alongside cone-frustum volume from the previous lesson.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                                   | Teacher Moves                                               | Sensemaking Strategy                                             | Formative Assessment Strategy                            |
|----------------------|----------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------|
| <b>Predict Phase</b> | Learners are shown a diagram of a frustum of a square pyramid (wider | Display the diagram of the mapunjas frustum on the board or | Think-Pair-Share Prediction: Learners first commit to individual | Observable indicator: Learners' written predictions show |

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| <p> <b>VIDEO:</b><br/> <a href="#">Surface area word problem example</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Pyramids Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                             | <p>square base, narrower square top, vertical height h) labelled as a traditional mapunjas grain store from the Rift Valley. They are asked: 'This frustum grain store has a square base of side 2 m, a square top of side 1 m, and a height of 1.5 m. Predict: do you think the formula for its volume will look the same as the cone-frustum formula we used in Lesson 7? Write your prediction and your reasoning in your exercise book. You have 3 minutes — work independently.' Learners write individual predictions, then share with a shoulder partner for 1 minute.</p>                                                                                                                                                                                                 | <p>projector. Say: 'Yesterday we found the volume of a cone frustum — the same shape used in those Kisumu storage tanks. Today we are looking at a different frustum: one cut from a pyramid, not a cone. Before we do any mathematics, I want you to think and predict.' Distribute the diagram handout. Set a 3-minute silent individual writing timer. After 3 minutes, say: 'Now turn to your shoulder partner and share your prediction. You have 1 minute.' After partner share, cold-call 3 learners: 'Wanjiku, what did you predict and why?' — 'Otieno, do you agree or disagree with Wanjiku? Why?' — 'Achieng, what was different about your reasoning?' Apply 10–15 seconds of WAIT TIME after each question before prompting. Record key prediction phrases on the board under the heading: 'Our Predictions — Pyramid Frustum Volume'. Do NOT confirm or correct predictions yet.</p> | <p>reasoning (reducing anchoring bias), then test their thinking against a peer, then hear three contrasting views publicly. This activates prior knowledge of cone frustums and surfaces misconceptions (e.g., 'the formula must use <math>\pi r^2</math>') before instruction begins.</p>                                                                  | <p>engagement with the structural analogy to cone frustums. Listen for whether learners correctly identify that the circular cross-section (<math>\pi r^2</math>) will be replaced by a square or rectangular area. Flag learners who predict a completely unrelated formula — they may need scaffolding during the derivation phase.</p>                                                                                    |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Surface area word problem example</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Surface Area and Volume of Pyramids (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners watch a YouTube video (pre-selected by the teacher) that demonstrates the derivation of the volume of a frustum of a pyramid by: (1) visualising the frustum as a large pyramid with its top cut off; (2) expressing the height of the full pyramid in terms of the frustum height and the scale factor of the two bases; (3) subtracting <math>V_{\text{small}}</math> from <math>V_{\text{large}}</math> to obtain <math>V_{\text{frustum}} = (h/3)(A_1 + A_2 + \sqrt{A_1 A_2})</math>. While watching, learners complete a structured note-taking frame: 'The frustum is formed by removing a _____ from a larger _____. The formula for the volume of a full pyramid is _____. The volume of the frustum = <math>V_{\text{large}} - V_{\text{small}}</math> =</p> | <p>Before playing the video, distribute the note-taking frame and say: 'As you watch, fill in this frame. Do not write everything — just the key mathematical steps. We will pause if needed.' Play the video. Pause at the moment the subtraction <math>V_{\text{large}} - V_{\text{small}}</math> is first written on screen and ask: 'Freeze — where have we seen this idea of subtracting two volumes before? Think for 10 seconds.' (WAIT TIME 10 seconds.) Cold-call 2 learners: 'Kamau, where did we subtract volumes in this unit?' — 'Nafula, can you connect that to what we just saw?' Resume the video to its conclusion. After the video, say: 'Compare your note-taking frame</p>                                                                                                                                                                                                     | <p>Structured Note-Taking Frame + Video Pause-and-Connect: The partially completed frame focuses learner attention on the logical structure of the derivation rather than passive watching. The mid-video pause explicitly connects the subtraction strategy to prior lessons, building a coherent mental schema across the evidence-gathering sequence.</p> | <p>Observable indicator: Completed note-taking frames show the correct formula with <math>A_1</math> and <math>A_2</math> correctly identified as the areas of the two parallel bases. Circulate during the tablemate comparison — learners who have left the <math>\sqrt{(A_1 A_2)}</math> term blank or written '<math>\pi r^2</math>' need immediate re-teaching of the geometric mean term before the Explain phase.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | _____' After the video, learners compare their notes with a tablemate for 2 minutes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | with your tablemate. Agree on the completed formula. You have 2 minutes.' Then write the formula $V = (h/3)(A_1 + A_2 + \sqrt{A_1A_2})$ on the board. Say: 'Let us unpack each term. What is $A_1$ ? What is $A_2$ ? What does $\sqrt{A_1A_2}$ represent geometrically?' Cold-call 3 learners for each term. Apply 12 seconds of WAIT TIME after each question.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Explain Phase</b><br> <b>VIDEO:</b><br><a href="#">Surface area word problem example</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Apply volume of solids</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners work through two Kenya-context application problems in pairs, then one learner from each pair presents a solution step to the class. Problem 1 (Agribusiness — Rift Valley): A frustum-shaped mapunjas grain store used by a maize farmer in Nakuru has a square base of side 3 m, a square top of side 1.5 m, and a vertical height of 2 m. Calculate the volume of grain it can hold in cubic metres, and convert to litres ( $1 \text{ m}^3 = 1\,000 \text{ L}$ ). Problem 2 (Construction — Siaya County): A frustum-shaped concrete base for a water tower in Siaya County has a rectangular base $4 \text{ m} \times 3 \text{ m}$ , a rectangular top $2 \text{ m} \times 1.5 \text{ m}$ , and a height of 2.5 m. Calculate its volume. After solving, learners write one sentence connecting their answer to the phenomenon: 'This tells us that the frustum of a pyramid shape...' | Say: 'We now apply our formula to two real structures in Kenya — one on a farm in Nakuru, one in Siaya County, very close to the Kisumu factory we have been investigating all term.' Display both problems on the board. Say: 'Work in pairs. For each problem, write out: (1) the formula, (2) $A_1$ and $A_2$ calculated, (3) substitution, (4) final answer with units. You have 12 minutes.' Circulate and target struggling pairs first. After 12 minutes, cold-call one learner to write Problem 1 Step 1 on the board, a different learner for Step 2, continuing until the solution is complete (minimum 4 different learners contribute). Apply 10 seconds of WAIT TIME before each cold-call. After Problem 1 is fully solved, repeat the process for Problem 2 with new learners. After both problems, say: 'Now individually, write one sentence starting with: This tells us that the frustum of a pyramid shape... connecting your answer to the Kisumu factory tanks. You have 2 minutes.' Cold-call 3 learners to read their sentences aloud. | Collaborative Problem Solving + Relay Presentation: Pairs reduce cognitive load during first application, while the relay presentation (each learner contributes one step publicly) distributes accountability and ensures every step of the formula is examined by the whole class. The phenomenon-connection sentence forces explicit transfer from abstract calculation to the anchoring context. | Observable indicator: Learners correctly compute $A_1$ and $A_2$ as areas of rectangular/square faces (not perimeters or volumes), correctly evaluate $\sqrt{A_1A_2}$ using a calculator, and produce answers with correct units ( $\text{m}^3$ and L). Flag pairs who multiply $h$ by $(A_1 + A_2)$ without the $\sqrt{A_1A_2}$ term — this is the most common structural error and requires targeted re-teaching of the geometric mean term. |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Learners return to the class Driving Question Board. Each learner receives one sticky note. They write a single evidence statement in the format: 'We now                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Say: 'We are now on Lesson 8 of 10. Our Driving Question Board has been growing all term. Today we add evidence about pyramid frustum volumes.' Distribute sticky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Evidence Accumulation on the DQB: The structured sentence frame ('We now know... which helps answer the driving question because...') requires learners to                                                                                                                                                                                                                                           | Observable indicator: Sticky notes contain a specific numerical value from today's lesson (not a vague statement) AND an explicit connection to the phenomenon                                                                                                                                                                                                                                                                                 |

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| <p><a href="#">Surface area word problem example</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Pyramids Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES</a> for similar readings</p>                                                        | <p>know that [evidence from today's lesson], which helps answer the driving question because [connection to material use / storage capacity].' Example: 'We now know that a pyramid frustum with base 3 m × 3 m, top 1.5 m × 1.5 m, and height 2 m holds 10.5 m<sup>3</sup> (10 500 L), which helps answer the driving question because a manufacturer can calculate exactly how much material to enclose that volume.' Learners post their sticky notes and the class does a 3-minute gallery read of five posted notes, with the teacher facilitating a brief whole-class synthesis.</p>                                                                                                                                                                                                                       | <p>notes. Say: 'Write your evidence statement using the sentence frame on the board. You have 3 minutes — write independently.' After 3 minutes, ask learners to post their notes on the DQB under the column heading: 'Volume Evidence'. Say: 'Quickly read five notes posted by your classmates. You have 3 minutes.' After the gallery read, cold-call 2 learners: 'Mwangi, which note surprised you most and why?' — 'Chebet, which note best connects volume to the Kisumu factory situation?' Apply 12 seconds of WAIT TIME. Summarise by pointing to the DQB and saying: 'Notice that we now have volume evidence for cones, cone-frustums, and pyramid-frustums. In Lessons 9 and 10 we will bring all of this together to fully answer our driving question about the Kisumu factory tanks.'</p> | <p>do two cognitive moves simultaneously — retrieve today's evidence AND connect it to the overarching question. The gallery read surfaces the range of evidence the class has collectively gathered, reinforcing that science and mathematics learning is cumulative.</p>                                                                                                                                                                                                       | <p>(material use, storage capacity, or manufacturing cost). Vague notes such as 'volume is important' indicate the learner has not yet made the phenomenon connection and should be followed up in Phase 5.</p>                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Surface area word problem example</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Pyramids Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES</a> for similar readings</p> | <p>Learners retrieve their cumulative model diagrams from Lesson 7. Their existing models show the cone-frustum and its volume formula. Today, learners add a new panel to their model: a clearly labelled diagram of a pyramid frustum showing dimensions <math>A_1</math> (larger base area), <math>A_2</math> (smaller base area), and <math>h</math> (vertical height), with the formula <math>V = (h/3)(A_1 + A_2 + \sqrt{(A_1 A_2)})</math> written below. Alongside the diagram, learners write two sentences: (1) How the pyramid-frustum volume formula is structurally similar to the cone-frustum formula from Lesson 7; (2) Which frustum shape (cone or pyramid) they predict would be chosen by the Kisumu factory manufacturer and why — this serves as a bridge prediction for Lessons 9–10.</p> | <p>Say: 'Take out your cumulative model from last lesson. You are going to add a new panel for pyramid frustums — this grows your model just like our understanding of the Kisumu factory tanks has grown.' Allow 8 minutes for learners to draw the labelled diagram and write both sentences. Circulate and prompt with: 'How does your pyramid-frustum panel connect to the cone-frustum panel you drew last lesson?' and 'If you were the manufacturer in Kisumu, which shape would you use for a 5 000-litre tank and why?' After 8 minutes, cold-call 3 learners to share their bridge prediction sentence: 'Odhiambo, which shape do you predict the manufacturer would choose?' — 'Auma, do you agree? What evidence from today</p>                                                               | <p>Cumulative Model Revision with Bridge Prediction: Adding a new panel to an existing model makes the growth of understanding visible and concrete. The bridge prediction (which shape would the manufacturer choose?) activates forward-looking reasoning, priming learners to look for comparative volume-surface area data in Lessons 9–10 rather than treating each lesson as isolated. This mirrors how engineers use iterative modelling in professional design work.</p> | <p>Observable indicator: The pyramid-frustum diagram is correctly labelled with <math>A_1</math>, <math>A_2</math>, and <math>h</math> (not <math>r</math>, not <math>l</math>). The formula is written correctly, including the <math>\sqrt{(A_1 A_2)}</math> term. The structural similarity sentence references the cone-frustum formula from Lesson 7 (indicating cross-lesson retrieval). The bridge prediction sentence references at least one specific mathematical property (volume, surface area, or dimensions) rather than a purely aesthetic or intuitive choice.</p> |

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|  |  | supports your prediction?' — 'Kariuki, what would you need to know to be certain?' Apply 10–12 seconds of WAIT TIME after each question. Close by saying: 'In Lessons 9 and 10, you will have all the evidence you need to fully answer the driving question. Make sure your model is complete and ready — it is your evidence record for the whole unit.' |  |
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| D. TEACHER REFLECTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <p>After the lesson, reflect on the following questions: (1) Did learners successfully transfer the subtraction-of-volumes strategy from cone frustums (Lesson 7) to pyramid frustums, or did they treat the new formula as an isolated memorisation task? If the latter, the connection between Lessons 7 and 8 needs to be made more explicit in the Observe phase next time. (2) Was the <math>\sqrt{(A_1A_2)}</math> geometric mean term understood conceptually, or only procedurally? If learners could apply it but not explain it, consider adding a brief visual showing <math>\sqrt{(A_1A_2)}</math> as the area of an 'intermediate' cross-section midway between the two bases. (3) Were the Kenya-context problems (Nakuru grain store, Siaya water tower base) sufficiently connected to the Kisumu factory phenomenon, or did learners treat them as separate unrelated word problems? Check DQB sticky notes — if fewer than 70% contain an explicit phenomenon connection, restructure the phenomenon-connection sentence prompt in Phase 3 for future lessons. (4) Did all learners contribute at least one public response during the relay presentation in Phase 3? Note names of learners who were not cold-called today and prioritise them in Lesson 9. (5) Are learners' cumulative models growing in sophistication across lessons, or are some learners starting fresh each time? Consider doing a brief model audit at the start of Lesson 9 before new content begins.</p> |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>What did I observe?</b>                                   | Learners observed a YouTube video demonstrating that a frustum of a pyramid can be formed by removing a smaller pyramid from a larger one, and that its volume is found by subtracting the two pyramid volumes — yielding the formula $V = (h/3)(A_1 + A_2 + \sqrt{(A_1A_2)})$ . They saw this applied to frustum-shaped structures including grain stores and construction bases in Kenya.                                                                               |
| <b>What did I learn?</b>                                     | Learners derived and applied the formula for the volume of a frustum of a pyramid: $V = (h/3)(A_1 + A_2 + \sqrt{(A_1A_2)})$ , where $A_1$ and $A_2$ are the areas of the two parallel bases and $h$ is the vertical height. They learned that this formula shares the same structural logic as the cone-frustum formula from Lesson 7, with circular cross-section areas ( $\pi r^2$ ) replaced by polygonal base areas.                                                  |
| <b>How does this explain the phenomenon?</b>                 | Learners explained that the shape of a frustum of a pyramid determines both how much it can hold (volume) and — when combined with surface area calculations — how much material is needed to build it. They connected this to the Kisumu factory phenomenon by reasoning that manufacturers choose between frustum shapes (cone-based vs. pyramid-based) based on comparative volume-to-surface-area efficiency, a question they will fully resolve in Lessons 9 and 10. |

## LESSON 9 (80 minutes): Volume of Composite Solids — Solving the Kisumu Tank Mystery

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Purpose</b>              | To enable learners to compute the volume of composite solids using a decomposition strategy, and to apply this knowledge alongside surface area understanding to fully explain the anchoring phenomenon — why a frustum tank uses less metal sheet than a composite cylinder-hemisphere tank of equal volume.                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Knowledge</b>            | Learners will know: (a) that composite solids are formed by combining two or more simple solids; (b) that the volume of a composite solid is found by decomposing it into recognisable simple solids and summing their individual volumes; (c) the volume formulas for cylinders, hemispheres, cones, frustums, and cuboids; (d) that volume and surface area are independent properties of a solid — the same volume can be enclosed by different amounts of surface area depending on shape.                                                                                                                                                                                                           |
| <b>Skills</b>               | Learners will be able to: (a) decompose a composite solid into its constituent simple solids; (b) correctly identify and apply the relevant volume formula for each constituent part; (c) calculate the total volume of composite solids such as cylinder + hemisphere and cone + cylinder; (d) compute side-by-side comparisons of surface area and volume for the two tank designs in the anchoring phenomenon; (e) interpret and communicate mathematical results in real-world engineering and manufacturing contexts.                                                                                                                                                                               |
| <b>Attitudes</b>            | Learners will appreciate: (a) the power of mathematical decomposition as a problem-solving strategy; (b) how geometry and measurement directly inform cost-effective design decisions made by Kenyan manufacturers and engineers; (c) the value of precision and attention to units when making decisions that affect communities in rural Kenya.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Key Inquiry Question</b> | How does the shape of a solid affect its surface area and volume — and how do manufacturers in Kenya use this knowledge to design tanks that minimise material cost while maximising storage capacity?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Purpose in Storyline</b> | This is the culminating evidence-gathering lesson (Lesson 9 of 10). Learners have previously computed surface areas and volumes of individual solids (cylinders, cones, frustums, spheres, hemispheres). Today they unite all prior evidence by: (1) using a YouTube video to master the decomposition strategy for composite solid volumes; (2) computing the volume of both tank designs from the Kisumu factory scenario; and (3) placing surface area and volume results side by side to fully explain why the frustum tank uses less material while holding the same volume. This lesson resolves the central tension of the anchoring phenomenon and prepares learners for the culminating lesson. |
| <b>Safety Notes</b>         | No physical hazards. Ensure all video content is pre-loaded on ARES offline platform before the lesson. Remind learners to handle any physical solid models carefully if used. Encourage respectful mathematical discourse — no ridicule of calculation errors.                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### B. LESSON OVERVIEW

In this 80-minute culminating evidence lesson, learners watch a short video on composite solid volumes, then apply a decomposition strategy to compute the volumes of the two Kisumu factory tank designs (cylinder + hemisphere, and frustum). Working in groups, they place their computed volumes alongside previously gathered surface area data to construct a full side-by-side comparison. Through guided sensemaking, learners articulate — for the first time with complete mathematical evidence — why the frustum tank uses less metal sheet despite holding the same 5,000 litres. The lesson closes with learners revising their cumulative models and updating the Driving Question Board with answers they can now confidently support with data.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                       | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Composite Solids Principles - Basic</a><br>Source: CK-12<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Volume and surface area word problems</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners are presented with a recap slide showing the two Kisumu factory tanks side by side — the composite cylinder-hemisphere tank and the frustum tank — both labelled as holding 5,000 litres. Each learner independently answers three predict questions in their exercise books: (1) 'How do you think we can find the volume of a solid made of two shapes joined together — like the cylinder-hemisphere tank?' (2) 'If both tanks hold exactly 5,000 litres, what does that tell you about their volumes?' (3) 'Do you think a solid with a larger volume always has a larger surface area? Explain your thinking.' Learners write their predictions without peer consultation, committing to an individual position before evidence is gathered. | Display the two tank diagrams on the board (or ARES screen). Say: 'Before we calculate anything, I want you to think independently. Look at these two tanks from the Kisumu factory. Both hold 5,000 litres. In your exercise books, answer the three questions on the board — you have 4 minutes, working alone.' [Allow 4 minutes of silent individual writing.] After time is up, use cold-call: select 4 learners (mix of confident and quieter voices) to share their prediction for Question 3. After each response, ask the class: 'Hands up if you agreed with that prediction.' Do NOT correct predictions yet. Record a tally of class predictions on the board under two columns: 'Volume always determines surface area' vs 'Volume and surface area are independent.' Say: 'We are going to find out who is right — and by the end of today you will be able to explain exactly what is happening at that Kisumu factory.' | Prediction Commitment Strategy — by writing individual predictions before instruction, learners activate prior knowledge, surface misconceptions (particularly the common conflation of volume and surface area), and create cognitive dissonance that the lesson will resolve. Recording the class tally makes the range of thinking visible and creates a public record to return to.                                                                    | Observable indicator: Are learners able to articulate — even informally — that volume and surface area might be independent? Listen for misconceptions such as 'bigger volume = bigger surface area always.' Identify 2–3 learners who express nuanced thinking (e.g., 'it depends on the shape') — these learners can be called on during the Explain phase to anchor the class discussion.                                                                                              |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Volume of triangular prism &amp; cube</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Solid geometry word problems</a><br>Source: Khan                                                                                                                                         | Learners watch a pre-loaded YouTube video (via ARES offline platform) on the volume of composite solids — approximately 6–8 minutes. The video demonstrates the decomposition strategy: identify the constituent simple solids, recall the volume formula for each, compute individually, then sum (or subtract where appropriate). After the video, learners complete a structured Video Notes template in their exercise books with four columns: (1) Composite Solid                                                                                                                                                                                                                                                                                    | Before playing the video, say: 'As you watch, your job is to be a detective. Fill in your Video Notes template. Pay special attention to the decomposition strategy — how does the presenter break the shape apart? You will use this exact strategy in the next 20 minutes.' [Play the video — approximately 6–8 minutes. Monitor that all learners are writing on their template, not passively watching.] After the video, pause and cold-call 3 learners: 'Tell me one solid from the video and how                                                                                                                                                                                                                                                                                                                                                                                                                                 | Decomposition Strategy + Structured Note-Taking — the video note template scaffolds active processing of the video rather than passive viewing. The structured format (decompose → formula → substitute → sum) mirrors the scientific practice of using mathematics and computational thinking (NGSS SEP 5) and builds transferable problem-solving habits. Immediate application to the Kisumu tanks ensures the video content is anchored to the driving | Observable indicator: During group work, check that each group has correctly written $V_{\text{total}} = V_{\text{cylinder}} + V_{\text{hemisphere}}$ for Tank 1, and is applying the frustum volume formula $V = (\pi h/3)(R^2 + Rr + r^2)$ for Tank 2. Listen for groups discussing unit conversion ( $1,000 \text{ cm}^3 = 1 \text{ litre}$ ). If any group skips the decomposition step and jumps to a formula, prompt: 'Show me on paper how you broke this shape into parts first.' |

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| <p>Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Shown, (2) Parts It Was Decomposed Into, (3) Volume Formula Used for Each Part, (4) How the Total Volume Was Found. Learners then move into their factory investigation groups (4–5 learners per group). Each group receives a data card with the dimensions of both Kisumu tank designs and the surface area values computed in previous lessons. Groups are tasked with computing the volume of each tank using the decomposition strategy modelled in the video.</p>                                                                                                                                                                                                                                                                                                                                                                              | <p>its volume was found.' Allow 10 seconds of wait time after each question before accepting a response. Then say: 'Now apply this strategy to the Kisumu tanks. Your data cards have all the dimensions. In your groups, calculate: (a) the volume of the cylinder-hemisphere tank, and (b) the volume of the frustum tank. Show every step — decompose, formula, substitute, calculate. You have 18 minutes.' Circulate and check: Are groups correctly decomposing? Are they using the right formula for the hemisphere [<math>V = (2/3)\pi r^3</math>]? Are they converting units correctly (cm<sup>3</sup> to litres)? Prompt struggling groups with: 'What two shapes make up that tank? Write them down first.'</p>                                                                                                                                                                | <p>phenomenon rather than treated as abstract content.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Composite Solids Principles - Basic</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Composite Solids (Flexbook)</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Groups compile their computed volumes onto a shared class Comparison Table on the board. The table has three columns: Tank Design   Surface Area (from previous lessons)   Volume (computed today). Once the table is complete with both tanks' data, learners engage in a guided whole-class discussion to interpret the table. They are asked: 'Both tanks hold 5,000 litres — do your calculated volumes confirm this?' and 'Look at the surface area column. Which tank has a smaller surface area? What does that mean for the manufacturer in Kisumu?' Learners then individually write a 'Mathematical Explanation' paragraph in their exercise books answering: 'Using the numbers in our comparison table, explain in full sentences why the frustum tank is more cost-effective to manufacture than the cylinder-hemisphere tank, even</p> | <p>Invite one group spokesperson per tank to present their volume calculation to the class. After each presentation, cold-call 2 different learners to verify: 'Do you agree with their decomposition? Thumbs up or thumbs down — and explain.' Allow 12 seconds wait time before accepting responses. Once the Comparison Table is complete on the board, say: 'Look at this table carefully. Both volumes should be approximately 5,000 litres — if yours are significantly different, check your unit conversion. Now — what story does this table tell us about the Kisumu factory?' [10 seconds wait time. Cold-call 3 learners from different ends of the room.] After discussion, say: 'Now I want each of you — individually — to write a full explanation paragraph. Use the actual numbers from our table. Mention surface area, volume, cost of metal sheet, and what this</p> | <p>Comparison Table + Evidence-Based Explanation Writing — the comparison table (NGSS SEP 6: Constructing Explanations) forces learners to treat mathematical results as evidence to be interpreted, not just numbers to compute. Writing the explanation paragraph develops scientific communication and connects abstract geometry to lived Kenyan economic realities. By anchoring the explanation in the Kisumu phenomenon, learners experience mathematics as a tool for understanding their world.</p> | <p>Observable indicator: In the written explanation paragraph, look for three elements: (1) explicit reference to the numerical surface area values showing the frustum has a smaller surface area; (2) a statement that both tanks have approximately equal volumes (~5,000 litres); (3) a logical connective linking these two facts to manufacturing cost (e.g., 'less metal sheet is needed, so the frustum tank costs less to produce'). Flag any learner who conflates volume and surface area in their paragraph for targeted follow-up in Lesson 10.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | though both hold the same volume of water.' They apply their knowledge of surface area (cost of material) vs. volume (storage capacity) using actual computed values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | means for a manufacturer trying to serve rural families in Siaya County at a lower price. You have 8 minutes.' Circulate and read over shoulders. Prompt with: 'You mentioned surface area is smaller — so what does that mean in shillings for the manufacturer?' and 'Why might a family in Siaya prefer the frustum tank?'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Composite Solids Principles - Basic</a><br>Source: CK-12<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Composite Solids Practice</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners return to the class Driving Question Board. In pairs, they review the questions and 'I wonder...' statements posted in earlier lessons. They identify which questions the today's comparison table now answers with mathematical evidence. Each pair writes a green sticky note (or uses the ARES DQB digital tool) with an 'Evidence-Based Answer' — beginning with the sentence stem: 'We can now explain that... because our calculations show...' They post their answers next to the relevant question on the DQB. Pairs also identify one new question that today's investigation has raised — for example: 'Are there other shapes that could hold 5,000 litres with even less surface area than the frustum?' and post it as a new 'I wonder...' note. | Direct learners to the DQB: 'Go back to our Driving Question Board. Today was a big day — we finally have both the surface area AND the volume of both tanks. Which questions on our board can we now answer with real numbers?' [Allow 3 minutes for pairs to review and write their green sticky note.] Cold-call 3 pairs to read their Evidence-Based Answer aloud. After each, ask the class: 'Does this answer fully explain the Kisumu phenomenon? What evidence supports it?' Then say: 'Even great investigators always find new questions. Post one new question your group is wondering about.' After posting, read 2–3 new questions aloud and say: 'These are excellent mathematical instincts. In Lesson 10 — our final lesson — we will bring everything together and you will design your own investigation to answer questions like these.' | Driving Question Board Evidence Mapping — returning to the DQB with colour-coded evidence answers (NGSS SEP 7: Engaging in Argument from Evidence) makes visible the progression of learners' understanding across the unit. The sentence stem 'We can now explain that... because our calculations show...' trains learners to distinguish claims from evidence-backed explanations. New 'I wonder' questions sustain curiosity and signal that mathematical investigation is generative, not terminal. | Observable indicator: Are pairs able to cite specific numerical evidence in their green sticky notes (e.g., surface area values, both volumes $\approx 5,000$ litres) rather than writing vague statements? A high-quality green note reads: 'We can now explain that the frustum tank uses less metal because its surface area is $[X]$ $\text{cm}^2$ compared to $[Y]$ $\text{cm}^2$ for the cylinder-hemisphere tank, even though both hold approximately 5,000 litres.' Flag pairs who write only general statements without numbers for reinforcement during Lesson 10. |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Composite Solids Principles - Basic</a><br>Source: CK-12<br> <a href="#">Search ARES for similar videos</a>                                                                                                                                                                                                                                                                                                                            | Learners retrieve their cumulative model from previous lessons — a labelled diagram of both tank designs showing surface area annotations. Today they add a new layer: they annotate each tank diagram with its total volume, clearly showing the decomposition (e.g., for the cylinder-hemisphere tank, a dotted line separating the                                                                                                                                                                                                                                                                                                                                                                                                                                   | Say: 'Take out your cumulative model from last lesson. Today we complete it. Add the volume annotations — show the decomposition with a dotted line and label each part's volume. Then write your Model Explanation box and your Design Verdict arrow. Your model should now tell the complete story of the Kisumu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Cumulative Model Revision (NGSS SEP 2: Developing and Using Models) — annotating the model with volume decomposition and the Design Verdict arrow transforms the diagram from a surface area reference tool into a complete engineering analysis artefact. The iterative model-building across all 9 lessons                                                                                                                                                                                             | Observable indicator: A complete and accurate model will show: (1) both tank diagrams with correct surface area labels (from prior lessons); (2) volume decomposition annotations with correct part-volumes summing to $\sim 5,000$ litres; (3) a written Model Explanation box referencing both surface area and volume; (4) a                                                                                                                                                                                                                                              |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Volume and surface area word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>cylinder body from the hemispherical lid, with the volume of each part labelled). Below their diagram, learners write a 'Model Explanation' box — two to three sentences summarising: (1) how the volume of each tank was calculated (decomposition strategy), (2) how the volumes compare, and (3) how the surface area and volume data together explain the manufacturer's design choice. They also add a 'Design Verdict' arrow pointing to the frustum tank, annotated: 'More cost-effective: same volume, less surface area → less metal sheet → lower manufacturing cost for Kisumu factory.'</p> | <p>factory investigation — shape, surface area, volume, and cost. You have 10 minutes.' Circulate and ask probing questions: 'If someone who had never heard of this investigation looked at your model, would they understand why the frustum is chosen? What would you add?' Cold-call 2 learners to display their completed models under the document camera (or hold up for class view). For each, invite peer feedback: 'One strength of this model, and one thing that could make it even clearer.' Close by saying: 'Your model now contains the full mathematical explanation for what those Grade 10 students observed at the Kisumu factory. You have moved from wondering to knowing — backed by evidence and calculation. In our final lesson, you will use this model to design and present your own engineering recommendation.'</p> | <p>mirrors how real engineers and scientists refine their understanding as new evidence accumulates. Peer feedback on model clarity builds communication and critical thinking competencies.</p> | <p>Design Verdict arrow correctly identifying the frustum as more cost-effective. Use this model as a pre-assessment artefact for the Lesson 10 culminating task — note which learners have all four elements and which need support.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were learners able to correctly apply the decomposition strategy to the composite cylinder-hemisphere tank, particularly using  $V = (2/3)\pi r^3$  for the hemisphere rather than the full sphere formula? Which groups struggled and why? (2) When learners wrote their Evidence-Based Explanation paragraphs, did they successfully distinguish between surface area (material cost) and volume (storage capacity)? Or did some still conflate the two concepts — and if so, how will you address this in Lesson 10's opening? (3) Did the Comparison Table on the board successfully create the 'aha moment' around the anchoring phenomenon? If not, what additional scaffolding (e.g., converting volumes to litres side by side, computing cost per litre of storage) might make the relationship more concrete? (4) How complete and accurate are learners' cumulative models? Are they ready to use these models as the basis for an engineering recommendation in Lesson 10? Identify 2–3 learners whose models are exemplary to share as anchor examples, and 2–3 learners who need targeted support at the start of Lesson 10.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                   |                                                                                                                                                                                                                                                                                                                                                            |
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| <p><b>What did I observe?</b></p> | <p>Learners watched a video on composite solid volumes and observed how any composite solid can be decomposed into simple constituent shapes (e.g., cylinder + hemisphere, cone + cylinder). They observed the decomposition strategy applied step by step, and then applied it themselves to compute the volumes of both Kisumu factory tank designs.</p> |
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| <b>What did I learn?</b>                     | Learners learned that: (a) the volume of a composite solid equals the sum of the volumes of its constituent parts; (b) the volume of the cylinder-hemisphere tank is $V = \pi r^2 h + (2/3)\pi r^3$ ; (c) the volume of the frustum tank is $V = (\pi h/3)(R^2 + Rr + r^2)$ ; (d) both tanks hold approximately 5,000 litres (equal volumes) but have different surface areas — confirming that volume and surface area are independent properties of a solid.                                                                                                                                                                            |
| <b>How does this explain the phenomenon?</b> | Learners explained — using a completed Comparison Table with actual computed values — that the frustum tank uses less metal sheet than the cylinder-hemisphere tank because it has a smaller surface area, even though both tanks hold exactly the same volume of water (5,000 litres). This explains the observation made by students at the Kisumu factory: shape determines how much material is needed (surface area) independently of how much water is stored (volume), and manufacturers strategically choose the frustum shape to reduce metal sheet consumption and lower production costs for rural households in Siaya County. |

## LESSON 10 (80 minutes): Real-Life Applications, Final Explanation and Cumulative Model Presentation

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>Purpose</b>              | To consolidate and apply knowledge of surface area and volume of solids by solving a rich real-life Kenyan engineering design task, presenting final annotated cumulative models, and completing the Driving Question Board with full explanations.                                                                                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | By the end of this lesson, the learner should be able to: (a) apply surface area and volume formulas for cubes, cuboids, cylinders, spheres, cones, pyramids, and composite/frustum solids to solve real-world design problems; (b) explain how the shape of a solid independently determines its surface area and volume; (c) justify the selection of an optimal solid shape for a given engineering or manufacturing context in Kenya.                                                                                                                                                                 |
| <b>Skills</b>               | By the end of this lesson, the learner should be able to: (a) solve multi-step surface area and volume problems set in Kenyan engineering contexts; (b) compare competing solid designs using calculated surface area and volume data; (c) construct, annotate, and present a final cumulative scientific model that traces understanding from Lesson 1 to Lesson 10; (d) communicate mathematical reasoning clearly using precise vocabulary.                                                                                                                                                            |
| <b>Attitudes</b>            | Learners appreciate the relevance of mathematics in real-world careers such as water engineering, architecture, and manufacturing in Kenya; demonstrate responsibility in the careful, accurate application of formulas; and show respect by listening to and building on peers' model presentations.                                                                                                                                                                                                                                                                                                     |
| <b>Key Inquiry Question</b> | 1. How do we apply surface area and volume of solids in day-to-day activities? 2. How does the shape of a solid affect its surface area and volume?                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose in Storyline</b> | This is the capstone lesson of the sub-strand. Learners return to the anchoring phenomenon — the Kisumu tank factory — and act as junior engineers tasked with recommending the most cost-efficient tank design for Siaya County School. Using all formulas and reasoning developed across Lessons 1–9, they calculate, compare, and justify their designs. They then present their final cumulative models, showing how their understanding has grown from naïve predictions in Lesson 1 to expert-level explanations today, and they close out every unanswered question on the Driving Question Board. |
| <b>Safety Notes</b>         | No physical hazards anticipated. If printed net cut-outs or model materials (cardboard, tape) are used during model presentation, remind learners to handle scissors carefully and keep workstations tidy. Ensure adequate space for group presentations to avoid crowding.                                                                                                                                                                                                                                                                                                                               |

### B. LESSON OVERVIEW

In this final lesson of Sub-Strand 2.4, learners act as junior engineers for a Kenyan water-tank manufacturer. They are presented with a rich design brief: recommend the most cost-efficient tank shape (from cylindrical-hemispherical composite, frustum, or pure cylinder) to supply 5,000 litres of water to Nyakach Secondary School in Siaya County, where metal sheet costs KSh 850 per square metre. Groups calculate surface area and volume for each competing design, compare costs, and present a justified recommendation. They then unveil their final annotated cumulative models, conduct a Lesson 1 vs. Lesson 10 model comparison, complete all remaining Driving Question Board entries, and reflect on careers in architecture, agriculture, manufacturing, and water engineering in Kenya that depend on this mathematics. Core Competencies addressed include Critical Thinking and Problem Solving, Communication and Collaboration, and Digital Literacy. PCIs addressed include Life Skills (decision-making) and Environmental Education (water conservation). Values addressed include Responsibility, Integrity, and Unity.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                      | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Cylinder volume &amp; surface area</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Volume and surface area of cylinders</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | <p>Learners are presented with the Final Design Brief projected on the board: 'You are a junior engineer at AquaFab Industries in Kisumu. Nyakach Secondary School in Siaya County needs a water storage tank that holds exactly 5,000 litres (5 m<sup>3</sup>). You must choose between three designs: Design A — a pure cylinder (r = 0.9 m, h = 1.97 m); Design B — a cylinder topped with a hemisphere (cylinder r = 0.85 m, h = 1.6 m, hemisphere r = 0.85 m); Design C — a frustum (R = 1.2 m, r = 0.7 m, h = 1.8 m). Metal sheet costs KSh 850 per m<sup>2</sup>. Which design will cost the school the least to build?' Working individually for 4 minutes, learners write their prediction — including which design they think uses the least material and WHY — in their exercise books. They then share predictions with a shoulder partner (Think-Pair). The teacher cold-calls 4 students to state predictions and reasoning aloud, recording key phrases on the board under 'Our Predictions' without affirming or correcting.</p> | <p>Display the Design Brief on the board. Say: 'Before we calculate anything, I want you to commit to a prediction. Write down: which design do you think uses the least metal sheet, and give me one reason based on what you have learned in this sub-strand.' Allow 4 minutes of silent individual thinking — do NOT circulate or answer questions during this time. After 4 minutes, say: 'Turn to your shoulder partner — you have 90 seconds to compare predictions. If you disagree, try to convince each other.' After pair discussion, cold-call exactly 4 students (aim for at least one from each design choice): 'Achieng, which design did you choose and why?', 'Otieno, do you agree or disagree with Achieng? Why?' Record key student phrases verbatim on the board. Do NOT evaluate: say 'Thank you — we will test that idea today.' Wait 10–12 seconds after each response before calling on the next learner.</p> | <p>Strategy: Commitment-and-Revisit Prediction. Learners commit to a specific, reasoned prediction before calculation, activating prior knowledge from Lessons 1–9 (formulas, shape comparisons, composite solids, frustums). Writing the prediction creates a cognitive anchor that will be revisited and potentially revised during Phase 3, making the final explanation more memorable and personally meaningful.</p> | <p>Observable indicator: Every learner has a written prediction with at least one mathematically grounded reason (e.g., references surface area formula, curved surface, slant height, composite shape). Teacher scans 5–6 books during the pair-share. Red flag: learners who write only 'Design C' with no mathematical reasoning — these learners need prompting during Phase 2 group work.</p>                                                                                |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Cylinder volume &amp; surface area</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Heights of Cylinders Given</a>                                                                                                                                                                                                 | <p>Learners work in their established groups of 4–5. Each group receives a printed AquaFab Design Task Card containing: (1) exact dimensions for all three designs; (2) a reminder formula sheet (cube, cuboid, cylinder, sphere, cone, frustum surface area and volume formulas); (3) the metal sheet cost rate (KSh 850/m<sup>2</sup>); (4) step-by-step scaffold questions: 'Step 1 — Verify each design holds <math>\geq 5 \text{ m}^3</math> (show</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Distribute Design Task Cards. Say: 'This is your engineering challenge. You are not just solving a maths problem — you are making a real recommendation that affects how much a school in Siaya County pays. Take this seriously.' Circulate continuously. At the 5-minute mark, check that all groups have correctly set up the volume verification for at least one design. Targeted prompt if stuck: 'What is the formula for the volume</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Strategy: Structured Engineering Design Task with Role Differentiation. By assigning calculations across group members and introducing a Quality Checker role, learners experience authentic scientific/engineering practice (NGSS SEP 5: Using Mathematics and Computational Thinking; SEP 6: Constructing Explanations). The real cost calculation (KSh 850/m<sup>2</sup>) ties abstract surface area values to</p>  | <p>Observable indicators: (1) Each group's comparison table is fully completed with correct units (m<sup>2</sup>, KSh). (2) Volume verification step shows <math>\geq 5 \text{ m}^3</math> for all three designs before cost comparison proceeds. (3) The Quality Checker can articulate at least one correction or confirmation. Red flags: groups who skip the volume verification step and go straight to surface area (they may be comparing tanks of different volumes —</p> |

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| <p><a href="#">Surface Area (Flexbook)</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>working). Step 2 — Calculate the total outer surface area for each design (excluding the base, since tanks sit on a platform). Step 3 — Calculate the metal sheet cost for each design. Step 4 — Record your results in the comparison table below.' Groups divide the three designs among members, then share results to complete the full table. Calculations are done on paper. Groups may use calculators. After 20 minutes, each group nominates a 'Quality Checker' who independently verifies at least one calculation per design before the group finalises their table. Key formulas used: Cylinder SA (lateral + top) = <math>2\pi rh + \pi r^2</math>; Hemisphere SA = <math>2\pi r^2</math>; Composite (B) = <math>2\pi r_c h_c + \pi r_c^2 + 2\pi r_h^2</math>; Frustum lateral SA = <math>\pi(R+r)l</math> where <math>l = \sqrt{h^2 + (R-r)^2}</math>; Frustum base area = <math>\pi R^2</math>. Volume checks: Cylinder V = <math>\pi r^2 h</math>; Composite V = <math>\pi r_c^2 h_c + (2/3)\pi r_h^3</math>; Frustum V = <math>(\pi h/3)(R^2 + Rr + r^2)</math>.</p> | <p>of a frustum? Check your formula sheet.' At the 12-minute mark, check that groups are computing lateral surface area only (not including base). Prompt: 'If the tank sits on a concrete platform, does the base need to be made of metal sheet? What does that mean for your surface area calculation?' At the 18-minute mark, ask the Quality Checker in 2 groups: 'Walk me through how your group calculated the slant height of the frustum. What values did you substitute?' Cold-call 2 Quality Checkers to narrate their verification process aloud to the class (30 seconds each). After 20 minutes, instruct groups to circle their recommended design in RED and write one sentence justifying their choice.</p> | <p>tangible economic consequences, directly connecting back to the phenomenon (the factory tour where students noticed the frustum used less metal).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>intervention: 'Before we compare costs, we must confirm both tanks hold the same amount. Show me your volume check. '); groups who include the base in the surface area (intervention: re-read the scenario together).</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Cylinder volume &amp; surface area</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/><a href="#">Heights of Cylinders Given Surface Area Practice</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p> | <p>Gallery Walk + Whole-Class Final Explanation. Each group posts their completed comparison table and recommended design on the wall or board. Groups rotate (5 minutes) to read other groups' tables, placing a sticky dot next to the recommendation they agree with most. After the gallery walk, learners return to seats. The teacher facilitates a 12-minute whole-class discussion structured around three Final Explanation questions displayed on the board: (Q1) 'What do our calculations show about the relationship between volume and surface area?' (Q2) 'Why does the frustum shape use less metal than the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>During the gallery walk, circulate and note which recommendation is most common. After learners return to seats, display the three Final Explanation questions. Ask Q1 first: 'Look at your tables. Surface area and volume are both properties of the same solid. What pattern do you notice about how they change across the three shapes?' Wait 12 seconds. Cold-call 3 students. Write key ideas on the board: 'Same volume <math>\neq</math> same surface area.' Ask Q2: 'Otieno's group says the frustum uses less metal. Can someone explain WHY — using the geometry of the shape — not just the numbers?' Wait 15 seconds. Cold-call 2</p>                                                                       | <p>Strategy: Gallery Walk + Before-and-After Reflection (NGSS SEP 6: Constructing Explanations; SEP 8: Obtaining, Evaluating, and Communicating Information). The gallery walk surfaces agreement and disagreement across groups, creating productive cognitive conflict before the whole-class discussion resolves it. The 'Before and After' reflection is a metacognitive tool that makes learning visible to the learner — they literally see their own growth from Lesson 1 naïve ideas to Lesson 10 expert explanations. This directly honours the KICD value of Responsibility (taking ownership of one's own learning</p> | <p>Observable indicators: (1) During the gallery walk, learners read and engage with other tables — not just their own. (2) 'Before and After' reflections explicitly name a formula or geometric property (not just 'I know more now'). (3) During Q2, at least one learner references slant height or the comparative radii of the frustum as geometric reasons for smaller surface area. Red flag: learners who write 'In Lesson 1 I thought the same as now' — these learners have not tracked their conceptual growth and need a one-on-one prompt: 'Go back to your Lesson 1 prediction card. Read exactly what you wrote. Has anything</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>composite cylinder-hemisphere for the same volume?' (Q3) 'How would a manufacturer in Kisumu use this information to reduce production costs?' Learners are then asked to return to their Lesson 1 prediction (recorded at the start of today AND in Lesson 1 of the sub-strand) and write a 'Before and After' reflection: 'In Lesson 1 I thought... Now I know... because...'. This is written individually. Finally, learners revisit their original Lesson 1 model of the phenomenon (the tank factory puzzle) and annotate it with all new understanding in a different colour.</p>                                                                                                                                                                                                                                                                | <p>students. Push for geometric reasoning: 'What about the slant height, the radii, the absence of a full circular top?' For Q3, say: 'Think like a business owner in Kisumu. How does knowing surface area formulas save money?' Cold-call 2 students. After discussion, say: 'Now I need 3 minutes of silence. Go back to your Lesson 1 prediction. Write: In Lesson 1 I thought... Now I know... because...'. After writing, invite 3 volunteers to read their reflection aloud. Celebrate growth explicitly: 'Notice how your explanation now includes the specific formula, the shape geometry, AND a real-world consequence. That is what expert mathematical thinking looks like.'</p>                                                                                                                                                           | <p>journey).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>changed?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Surface Area and Volume of Cylinders Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>The class gathers around the Driving Question Board that has been built up across all 10 lessons. The teacher reads aloud each remaining unanswered question (marked in YELLOW sticky notes from previous lessons). For each, the class nominates a 'Question Owner' — a student who feels they can now answer it — who writes a concise answer on a GREEN sticky note and places it over the yellow. The two KICD Key Inquiry Questions are addressed last and most formally: (KIQ 1) 'How do we apply surface area and volume of solids in day-to-day activities?' — the class co-constructs a list of 5 real Kenyan applications on the board (e.g., water tank design in Siaya, grain silo sizing in Eldoret, packaging design for Bidco Kenya, football production in Athi River, architectural dome design in Nairobi). (KIQ 2) 'How does the</p> | <p>Point to the DQB. Say: 'This board has been our learning journey for 10 lessons. Today we close it out. Every yellow question gets a green answer.' Read each yellow question aloud. Ask: 'Who owns this question? Who can answer it now?' Wait 8 seconds. If no volunteer, cold-call a student who showed understanding in Phase 3. After all yellow questions are answered, say: 'Now the two big questions that started everything.' Read KIQ 1. Say: 'Give me real Kenyan examples — industries, farms, buildings. Turn to your partner for 1 minute.' Cold-call 5 pairs to contribute one example each — compile the list on the board. For KIQ 2, say: 'In one sentence — not a formula, a sentence — how does shape affect surface area and volume?' Wait 15 seconds. Take 3 responses, synthesise them, and write the consensus sentence</p> | <p>Strategy: Collaborative DQB Closure (NGSS SEP 8: Obtaining, Evaluating, and Communicating Information). Closing the DQB is a visible, public record of the class's collective intellectual journey. Assigning 'Question Owners' gives individual learners personal accountability for a piece of the shared understanding. The co-written consensus answers model scientific communication — precise, evidence-based, collectively verified — directly mirroring how Kenyan engineers and manufacturers document and justify design decisions.</p> | <p>Observable indicators: (1) Every yellow sticky note on the DQB is covered by a green answer by the end of this phase. (2) The KIQ 2 consensus sentence includes at minimum: reference to shape, surface area, volume, and independence of these two quantities. (3) The phenomenon final explanation includes: the word 'frustum', a reference to slant height or comparative radii, and the phrase 'same volume'. Red flag: vague answers like 'shapes are different so surface area is different' — prompt: 'Can you be more specific? What property of the frustum's geometry leads to a smaller surface area?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>shape of a solid affect its surface area and volume?' — the class agrees on a one-sentence consensus answer that is written on a large card and pinned to the board. Finally, the class revisits the anchoring phenomenon question: 'Why does the frustum tank use less metal than the composite tank for the same 5,000-litre volume?' and co-writes the full, final explanation on a display card.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>aloud as you speak. For the phenomenon, say: 'We started with a puzzle in a factory in Kisumu. Can we now fully explain it? Auma, start us off.' Build the explanation collaboratively, adding each student's contribution. Read the final explanation aloud as a class.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Cylinder volume &amp; surface area</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Apply volume of solids</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Each group presents their final annotated cumulative model (built and revised across Lessons 1–10) to the class in a 2-minute presentation. The model must show: (1) a labelled diagram of at least two solid shapes with formulas; (2) annotations tracing how their understanding changed from Lesson 1 (e.g., 'In Lesson 1 we only knew the cylinder formula — arrow pointing to later annotation: By Lesson 6 we added the frustum'); (3) a final conclusion statement answering the Driving Question. After all groups present, the teacher facilitates a whole-class Careers in Mathematics reflection (5 minutes): learners brainstorm and discuss real Kenyan careers where surface area and volume knowledge is essential — water engineers at Nairobi City Water, architects at firms designing Nairobi's affordable housing projects, grain silo engineers in Eldoret, packaging designers at Bidco Industries, agricultural engineers advising tea farmers in Kericho on fertiliser storage, and construction managers for the Lamu Port project. Lesson closes with each learner writing one personal Exit</p> | <p>Introduce the model presentations: 'You have been building these models since Lesson 1. In 2 minutes, show the class what you now know and how your thinking changed.' Keep strict time (use a visible timer). After each presentation, ask one question to the presenting group — rotate these: 'What was the biggest change to your model between Lesson 1 and today?'; 'Which formula took the longest to understand and why?'; 'If you were advising the factory in Kisumu, what shape would you recommend and why?' After all groups, say: 'Let's talk about who actually uses this mathematics in Kenya.' Pose: 'Name a Kenyan career — not a general one — a specific type of professional in a specific Kenyan industry — who must calculate surface area or volume.' Wait 10 seconds. Cold-call 6 students. Add each career to a 'Careers Wall' on the board. Close with: 'Take 3 minutes — write your Exit Ticket. One sentence: the most important thing you learned and how you could use it in a future career.' Collect Exit Tickets at the door as learners leave.</p> | <p>Strategy: Cumulative Model Presentation with Careers Connect (NGSS SEP 2: Developing and Using Models; SEP 8: Communicating Information). Public model presentations consolidate individual and group understanding while exposing the whole class to multiple valid representations of the same mathematical concepts. The Careers Connect activity contextualises the mathematics within Kenya's economic and industrial landscape, fulfilling the KICD mandate that learning is connected to real life and future pathways. The Exit Ticket provides individual summative evidence of each learner's final understanding.</p> | <p>Observable indicators: (1) Each group's model contains at minimum two labelled solid shapes with correct formulas and at least two 'revision annotations' showing knowledge growth. (2) The Driving Question is answered in the model's conclusion statement using mathematical vocabulary (surface area, volume, shape, formula, cost). (3) Exit Tickets name a specific concept AND a specific career or application — not generic statements. Red flag: models that look the same as Lesson 1 (no visible revision or annotation) — prompt: 'Show me where you changed your model after Lesson 4. Where did you add the cone formula? Where did you correct something?' Red flag: Exit Tickets that say only 'I learned surface area' — return the card and ask the learner to add: '...of a [specific shape] using the formula [formula] and I could use this in [career/application].'</p> |

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|  | Ticket sentence: 'The most important thing I learned in this sub-strand, and how I could use it in a future career, is...' |  |  |  |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. **CONCEPTUAL GROWTH:** When learners read their 'In Lesson 1 I thought... Now I know...' reflections aloud, did their explanations demonstrate genuine conceptual growth — specifically, could they articulate WHY shape affects surface area independently of volume, using geometric reasoning (not just formula recall)?
2. **DRIVING QUESTION:** Was the class able to co-write a complete, precise final explanation of the anchoring phenomenon (the Kisumu tank factory puzzle) that referenced the frustum's geometry, slant height, and the independence of surface area and volume? If not, which specific misconception persisted — and how will you address it before the end-of-strand assessment?
3. **MODEL QUALITY:** Did the cumulative models show genuine revision across the 10 lessons? If most groups' models looked similar to their Lesson 1 versions with minimal annotation, reconsider whether model-building check-ins in the middle lessons (Lessons 4–7) were frequent and structured enough. Plan to build in a formal 'model revision checkpoint' in the middle of each future sub-strand.
4. **ENGAGEMENT AND EQUITY:** Did all learners — including quieter learners and those who struggled with formulas in earlier lessons — have a visible, valued role in the final presentations and DQB closure? Review your cold-call notes: did you call on the same 4–5 students throughout the lesson? Intentionally broaden participation in future lessons.
5. **CAREERS CONNECT:** Were learners able to name specific Kenyan careers (not just 'engineer' or 'architect' but 'water engineer at Nairobi City Water' or 'packaging designer at Bidco')? If responses were generic, consider inviting a local professional (e.g., a civil engineer from a Siaya County Water project) as a guest speaker or video call in a future lesson cycle.
6. **ASSESSMENT READINESS:** Review Exit Tickets before the next lesson. Identify learners who: (a) correctly named formula + shape + career (ready for assessment); (b) named formula but not career (need application practice); (c) wrote vague statements (need individual follow-up before the sub-strand summative assessment). Use this data to inform groupings for any remediation or extension tasks.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>What did I observe?</b> | Learners observed that three different solid shapes (a pure cylinder, a composite cylinder-hemisphere, and a frustum) can each hold exactly 5,000 litres (5 m <sup>3</sup> ) of water while having significantly different outer surface areas. By calculating and comparing the surface area and associated metal sheet cost (at KSh 850/m <sup>2</sup> ) for each design in the AquaFab Industries engineering task, learners directly observed that the frustum design produces the lowest surface area — and therefore the lowest manufacturing cost — for the same storage volume. They also observed, through the Gallery Walk, that different groups reached the same mathematical conclusion via different calculation pathways. |
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| <b>What did I learn?</b>                     | Learners learned that surface area and volume are independent properties of a solid — a shape can be redesigned to hold the same volume while using less surface material by changing its geometry. They learned and applied the full set of surface area and volume formulas for cylinders, hemispheres, composite solids, and frustums in a multi-step real-world engineering context. They learned that manufacturers and engineers in Kenya strategically select solid shapes to minimise material cost while meeting capacity requirements, and they connected this mathematics to careers in water engineering, architecture, packaging design, and agricultural storage across Kenya.                                           |
| <b>How does this explain the phenomenon?</b> | Learners can now fully explain the anchoring phenomenon: the frustum-shaped tank at the Kisumu factory uses less metal sheet than the composite cylinder-hemisphere tank for the same 5,000-litre volume because the frustum's geometry — specifically its sloping lateral surface, defined by slant height $l = \sqrt{h^2 + (R-r)^2}$ , and the fact that its top opening is narrower (smaller $r$ ) than a full cylinder's top — results in a smaller total outer surface area. Volume and surface area are determined by different aspects of a solid's shape and dimensions; they do not change proportionally together. Engineers exploit this geometric independence to design cost-efficient, high-capacity storage structures. |

| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
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| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |